SET 66 • HSG-GPE.A.2 • EXPRESSING GEOMETRIC PROPERTIES WITH EQUATIONS

Derive the equation of a parabola from a focus and directrix

MAIN	Geometry main question	**HSG-GPE.A.2**
BACKWARD 1	Distance between two points	**8.G.B.8**
BACKWARD 2	Distance from a point to a horizontal line	**6.NS.C.8**
BACKWARD 3	Points equidistant from two objects	**HSG-CO.C.9**
BACKWARD 4	Expand and simplify squares	**HSA-APR.A.1**
FORWARD 1	Find the focus and directrix from an equation	**HSG-GPE.A.2**
FORWARD 2	Design with the reflective property of parabolas	**HSG-MG.A.3**

SET 66 • HSG-GPE.A.2

MAIN

Geometry main question • High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.2

Derive the equation of the parabola with focus (0, 2) and directrix $y = -2$.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-M-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.2 • Set 66 • MAIN • Q1/5

Write the equation of the parabola with focus $(0, -3)$ and directrix $y = 3$.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-M-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.2 • Set 66 • MAIN • Q2/5

Write the equation of the parabola with focus (2, 5) and directrix y = 1.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-M-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.2 • Set 66 • MAIN • Q3/5

Write the equation of the parabola with focus (3, 0) and directrix $x = -3$.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-M-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.2 • Set 66 • MAIN • Q4/5

A parabola has focus (1, 4) and directrix y = 0. Its vertex is

A. (1, 4)

B. (0, 2)

C. (1, 0)

D. (1, 2)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-M-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.2 • Set 66 • MAIN • Q5/5

SET 66 • HSG-GPE.A.2

BACKWARD 1

Distance between two points • Grade 8 • Geometry • 8.G.B.8

Find the distance from (3, 6) to (0, 2).

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-B1-Q1

Grade 8 • Geometry • 8.G.B.8 • Set 66 • BACKWARD 1 • Q1/5

Write the distance from (x, y) to (0, 2) as an expression.

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-B1-Q2

Grade 8 • Geometry • 8.G.B.8 • Set 66 • BACKWARD 1 • Q2/5

The distance from (4, 1) to (1, 5) is

A. 7

B. 5

C. $\sqrt{7}$

D. 25

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-B1-Q3

Grade 8 • Geometry • 8.G.B.8 • Set 66 • BACKWARD 1 • Q3/5

True or false?

The distance formula comes from the Pythagorean Theorem.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-B1-Q4

Grade 8 • Geometry • 8.G.B.8 • Set 66 • BACKWARD 1 • Q4/5

True or false?

$\sqrt{(9 + 16)} = 3 + 4$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-B1-Q5

Grade 8 • Geometry • 8.G.B.8 • Set 66 • BACKWARD 1 • Q5/5

SET 66 • HSG-GPE.A.2

BACKWARD 2

Distance from a point to a horizontal line • Grade 6 • The Number System • 6.NS.C.8

How far is (5, 7) from the line y = 2?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-B2-Q1

Grade 6 • The Number System • 6.NS.C.8 • Set 66 • BACKWARD 2 • Q1/5

How far is (1, −3) from the line y = 4?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-B2-Q2

Grade 6 • The Number System • 6.NS.C.8 • Set 66 • BACKWARD 2 • Q2/5

The distance from (x, y) to $y = -2$ is

A. $|x + 2|$

B. y

C. $|y - 2|$

D. $|y + 2|$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-B2-Q3

Grade 6 • The Number System • 6.NS.C.8 • Set 66 • BACKWARD 2 • Q3/5

True or false?

The distance from a point to a horizontal line is measured straight up or down.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-B2-Q4

Grade 6 • The Number System • 6.NS.C.8 • Set 66 • BACKWARD 2 • Q4/5

True or false?

The distance from (6, 1) to y = 4 is 6.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-B2-Q5

Grade 6 • The Number System • 6.NS.C.8 • Set 66 • BACKWARD 2 • Q5/5

SET 66 • HSG-GPE.A.2

BACKWARD 3

Points equidistant from two objects • High School — Geometry • Geometry: Congruence • HSG-CO.C.9

True or false?

The set of points equidistant from two points is the perpendicular bisector of the segment joining them.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-B3-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 66 • BACKWARD 3 • Q1/5

A parabola is the set of points equidistant from what two objects?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-B3-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 66 • BACKWARD 3 • Q2/5

The vertex of a parabola is

A. at the focus

B. twice as far from the focus

C. halfway between the focus and the directrix

D. on the directrix

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-B3-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 66 • BACKWARD 3 • Q3/5

Focus (0, 6), directrix y = 0. Find the vertex.

Vertex: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-B3-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 66 • BACKWARD 3 • Q4/5

True or false?

Every point of a parabola is closer to the focus than to the directrix.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-B3-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 66 • BACKWARD 3 • Q5/5

SET 66 • HSG-GPE.A.2

BACKWARD 4

Expand and simplify squares • High School — Algebra I (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1

Expand $(y - 2)^2$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-B4-Q1

High School — Algebra I (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions HSA-APR.A.1 • Set 66 • BACKWARD 4 • Q1/5

Simplify $(y + 2)^2 - (y - 2)^2$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-B4-Q2

High School — Algebra I (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 66 • BACKWARD 4 • Q2/5

$y^2 + 6y + 9 - (y^2 - 6y + 9) =$

A. 18

B. $2y^2$

C. 0

D. $12y$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S66-B4-Q3**

High School — Algebra I (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 66 • BACKWARD 4 • Q3/5

True or false?

$(y + 3)^2 = y^2 + 6y + 9$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-B4-Q4

High School — Algebra I (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 66 • BACKWARD 4 • Q4/5

True or false?

$$(y - 2)^2 - y^2 = 4$$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-B4-Q5

High School — Algebra I (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 66 • BACKWARD 4 • Q5/5

SET 66 • HSG-GPE.A.2

FORWARD 1

Find the focus and directrix from an equation • High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.2

Find the focus and directrix of $y = x^2/12$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-F1-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.2 • Set 66 • FORWARD 1 • Q1/5

Find the vertex, focus, and directrix of $(x - 1)^2 = -8(y + 2)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-F1-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.2 • Set 66 • FORWARD 1 • Q2/5

True or false?

In $x^2 = 4py$, the focus is p units from the vertex.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-F1-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.2 • Set 66 • FORWARD 1 • Q3/5

The parabola $y^2 = -20x$ opens

A. left

B. up

C. down

D. right

Find the focus of $y = 2x^2$.

Focus: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-F1-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.2 • Set 66 • FORWARD 1 • Q5/5

SET 66 • HSG-GPE.A.2

FORWARD 2

Design with the reflective property of parabolas • High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3

A satellite dish is 4 ft across and 1 ft deep. How far from the vertex should the receiver (at the focus) be placed?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-F2-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 66 • FORWARD 2 • Q1/5

A headlight reflector has equation $x^2 = 12y$ (inches). Where should the bulb go?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-F2-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 66 • FORWARD 2 • Q2/5

True or false?

Rays parallel to the axis of a parabolic mirror reflect through the vertex.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-F2-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 66 • FORWARD 2 • Q3/5

A dish has focus 2 m above its vertex. Its equation (vertex at the origin) is

A. $x^2 = 4y$

B. $x^2 = 8y$

C. $y = 8x^2$

D. $x^2 = 2y$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-F2-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 66 • FORWARD 2 • Q4/5

A dish has equation $x^2 = 16y$ (meters) and is 8 m across. How deep is it?

Depth: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-F2-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 66 • FORWARD 2 • Q5/5

SET 67 • HSG-GPE.B.4 • EXPRESSING GEOMETRIC PROPERTIES WITH EQUATIONS

Prove or disprove the type of a quadrilateral with coordinates

MAIN	Geometry main question	**HSG-GPE.B.4**
BACKWARD 1	Find the slope between two points	**8.F.B.4**
BACKWARD 2	Find the distance between two points	**8.G.B.8**
BACKWARD 3	Properties that define special quadrilaterals	**HSG-CO.C.11**
FORWARD 1	Coordinate proofs with variables	**HSG-GPE.B.4**
FORWARD 2	Use the classification to find area	**HSG-GPE.B.7**

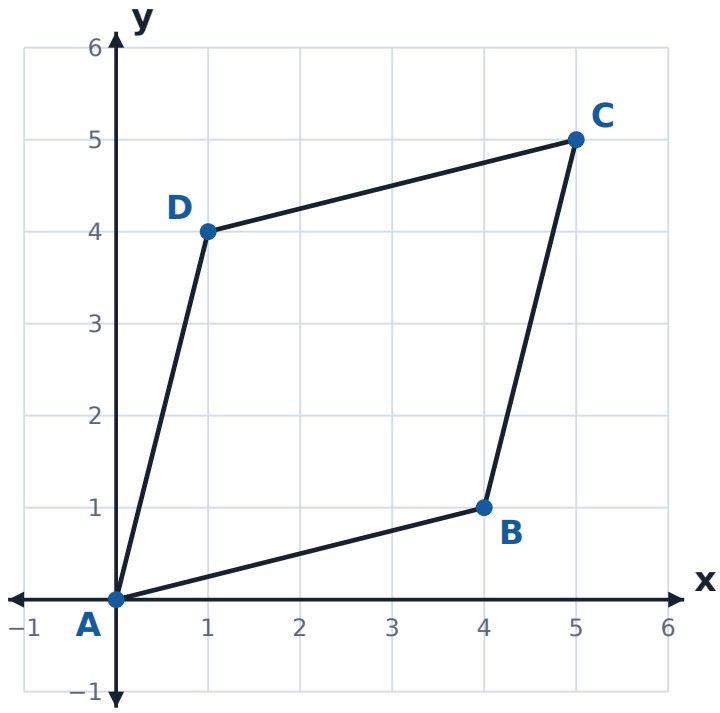
SET 67 • HSG-GPE.B.4

MAIN

Geometry main question • High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4

Prove that ABCD with A(0, 0), B(4, 1), C(5, 5), D(1, 4) is a rhombus.

Work space



Conclusion: _____

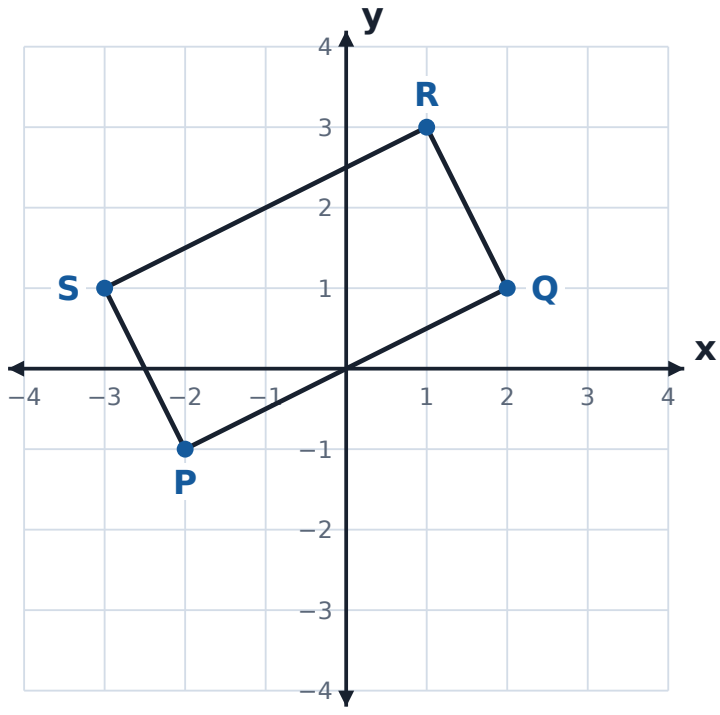
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-M-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 67 • MAIN • Q1/5

Prove that PQRS with P(−2, −1), Q(2, 1), R(1, 3), S(−3, 1) is a rectangle. Is it a square?

Work space



Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-M-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 67 • MAIN • Q2/5

Prove or disprove: KLMN with K(0, 0), L(5, 0), M(6, 3), N(0, 3) is a parallelogram.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-M-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 67 • MAIN • Q3/5

Classify the triangle with vertices A(1, 1), B(4, 5), C(8, 2) by its sides and angles. Justify.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-M-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 67 • MAIN • Q4/5

A quadrilateral is shown to be a parallelogram. Which additional calculation proves it is a rectangle?

A. Two adjacent sides have slopes whose product is -1

B. Two opposite sides have equal slopes

C. The diagonals have equal slopes

D. Two adjacent sides have equal length

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-M-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 67 • MAIN • Q5/5

SET 67 • HSG-GPE.B.4

BACKWARD 1

Find the slope between two points • Grade 8 • Functions • 8.F.B.4

Find the slope of the line through (1, 2) and (5, 4).

Slope: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-B1-Q1

Grade 8 • Functions • 8.F.B.4 • Set 67 • BACKWARD 1 • Q1/5

Find the slope of the line through (0, 3) and (2, −1).

Slope: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-B1-Q2

Grade 8 • Functions • 8.F.B.4 • Set 67 • BACKWARD 1 • Q2/5

The slope through (2, 5) and (6, 5) is

A. undefined

B. 0

C. 5

D. 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-B1-Q3

Grade 8 • Functions • 8.F.B.4 • Set 67 • BACKWARD 1 • Q3/5

True or false?

A vertical segment has an undefined slope.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-B1-Q4

Grade 8 • Functions • 8.F.B.4 • Set 67 • BACKWARD 1 • Q4/5

True or false?

Slope is run divided by rise.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-B1-Q5

Grade 8 • Functions • 8.F.B.4 • Set 67 • BACKWARD 1 • Q5/5

SET 67 • HSG-GPE.B.4

BACKWARD 2

Find the distance between two points • Grade 8 • Geometry • 8.G.B.8

Find the distance between (0, 0) and (4, 1).

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-B2-Q1

Grade 8 • Geometry • 8.G.B.8 • Set 67 • BACKWARD 2 • Q1/5

Find the distance between (1, 1) and (4, 5).

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-B2-Q2

Grade 8 • Geometry • 8.G.B.8 • Set 67 • BACKWARD 2 • Q2/5

Which pair of points is 10 apart?

A. (2, 3) and (8, 3)

B. (0, 0) and (6, 8)

C. (0, 0) and (10, 10)

D. (1, 1) and (5, 7)

True or false?

Two segments with the same length are congruent.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-B2-Q4

Grade 8 • Geometry • 8.G.B.8 • Set 67 • BACKWARD 2 • Q4/5

True or false?

$\sqrt{10} + \sqrt{10} = \sqrt{20}$

A. True

B. False

SET 67 • HSG-GPE.B.4

BACKWARD 3

Properties that define special quadrilaterals • High School — Geometry • Geometry: Congruence • HSG-CO.C.11

Which quadrilateral has four congruent sides?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-B3-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 67 • BACKWARD 3 • Q1/5

A parallelogram with one right angle is a ___.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-B3-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 67 • BACKWARD 3 • Q2/5

Which shows a quadrilateral is a parallelogram?

A. One pair of sides is parallel

B. Two adjacent sides are equal

C. One right angle

D. Both pairs of opposite sides are parallel

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-B3-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 67 • BACKWARD 3 • Q3/5

True or false?

Every square is a rhombus.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-B3-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 67 • BACKWARD 3 • Q4/5

True or false?

Every rhombus is a square.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-B3-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 67 • BACKWARD 3 • Q5/5

SET 67 • HSG-GPE.B.4

FORWARD 1

Coordinate proofs with variables • High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4

A rectangle has vertices O(0, 0), A(a, 0), B(a, b), C(0, b). Prove that its diagonals are congruent.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-F1-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 67 • FORWARD 1 • Q1/5

Triangle with vertices O(0, 0), A(2a, 0), B(2b, 2c). M and N are the midpoints of OB and AB. Prove that MN ∥ OA and MN = ½OA.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-F1-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 67 • FORWARD 1 • Q2/5

True or false?

Placing a figure with a vertex at the origin and a side on the x-axis is allowed in a coordinate proof.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-F1-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 67 • FORWARD 1 • Q3/5

Why do coordinates like (2a, 0) and (2b, 2c) make the midpoint proof easier?

A. The midpoints have no fractions

B. They make the triangle isosceles

C. They make a right angle

D. They make the slopes 0

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-F1-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 67 • FORWARD 1 • Q4/5

Why is a proof using A(6, 0) and B(2, 4) not enough to prove the midsegment theorem?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-F1-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 67 • FORWARD 1 • Q5/5

SET 67 • HSG-GPE.B.4

FORWARD 2

Use the classification to find area • High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7

ABCD with A(0, 0), B(4, 1), C(5, 5), D(1, 4) is a rhombus. Find its area using its diagonals.

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-F2-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 67 • FORWARD 2 • Q1/5

Rectangle PQRS has P(−2, −1), Q(2, 1), R(1, 3), S(−3, 1). Find its area.

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-F2-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 67 • FORWARD 2 • Q2/5

True or false?

Once a quadrilateral is proved a rhombus, its area is the product of two adjacent side lengths.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-F2-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 67 • FORWARD 2 • Q3/5

A square has vertices (0, 0), (3, 1), (2, 4), (−1, 3). Its area is

A. 16

B. √10

C. 10

D. 9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-F2-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 67 • FORWARD 2 • Q4/5

Explain why proving the shape first matters before using a special area formula.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-F2-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 67 • FORWARD 2 • Q5/5

Prove or disprove that a point lies on a circle, and other coordinate proofs

MAIN	Geometry main question	**HSG-GPE.B.4**
BACKWARD 1	Distance between two points	**8.G.B.8**
BACKWARD 2	The equation of a circle	**HSG-GPE.A.1**
BACKWARD 3	Substitute values into an expression	**6.EE.A.2**
FORWARD 1	Coordinate proofs about circles with variables	**HSG-GPE.B.4**
FORWARD 2	Find the tangent line to a circle at a point	**HSG-GPE.B.5**

SET 68 • HSG-GPE.B.4

MAIN

Geometry main question • High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4

A circle is centered at the origin and contains the point (0, 2). Prove or disprove that the point (1, √3) lies on the circle.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-M-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 68 • MAIN • Q1/5

Prove or disprove that (5, 2) lies on the circle with center (1, −1) and radius 5.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-M-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 68 • MAIN • Q2/5

Prove or disprove that $(-3, 6)$ lies on the circle $(x - 1)^2 + (y - 2)^2 = 30$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-M-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 68 • MAIN • Q3/5

Right triangle ABC has A(0, 0), B(8, 0), C(0, 6). Prove that the midpoint M of hypotenuse BC is equidistant from A, B, and C.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-M-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 68 • MAIN • Q4/5

The point $(4, -3)$ is ___ the circle $x^2 + y^2 = 20$.

A. on

B. inside

C. outside

D. the center of

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-M-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 68 • MAIN • Q5/5

SET 68 • HSG-GPE.B.4

BACKWARD 1

Distance between two points • Grade 8 • Geometry • 8.G.B.8

Distance from $(1, -1)$ to $(5, 2)$?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-B1-Q1

Grade 8 • Geometry • 8.G.B.8 • Set 68 • BACKWARD 1 • Q1/5

Distance from (0, 0) to (1, $\sqrt{3}$)?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-B1-Q2

Grade 8 • Geometry • 8.G.B.8 • Set 68 • BACKWARD 1 • Q2/5

Distance from (0, 0) to (−3, 4) is

A. 5

B. 1

C. 7

D. 25

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-B1-Q3

Grade 8 • Geometry • 8.G.B.8 • Set 68 • BACKWARD 1 • Q3/5

True or false?

$(-3)^2 = 9$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-B1-Q4

Grade 8 • Geometry • 8.G.B.8 • Set 68 • BACKWARD 1 • Q4/5

True or false?

The distance from (2, 2) to (2, 7) is 9.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-B1-Q5

Grade 8 • Geometry • 8.G.B.8 • Set 68 • BACKWARD 1 • Q5/5

SET 68 • HSG-GPE.B.4

BACKWARD 2

The equation of a circle • High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1

Write the equation of the circle with center $(1, -1)$ and radius 5.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-B2-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 68 • BACKWARD 2 • Q1/5

What is the radius of $(x - 1)^2 + (y - 2)^2 = 30$?

Radius: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-B2-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 68 • BACKWARD 2 • Q2/5

Center of $(x + 4)^2 + y^2 = 9$?

A. $(4, 0)$

B. $(-4, 0)$

C. $(-4, 9)$

D. $(0, -4)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-B2-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 68 • BACKWARD 2 • Q3/5

True or false?

A point satisfies a circle's equation exactly when it is on the circle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-B2-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 68 • BACKWARD 2 • Q4/5

True or false?

The radius of $x^2 + y^2 = 20$ is 20.

A. True

B. False

SET 68 • HSG-GPE.B.4

BACKWARD 3

Substitute values into an expression • Grade 6 • Expressions & Equations • 6.EE.A.2

Evaluate $(x - 1)^2 + (y - 2)^2$ when $x = -3$ and $y = 6$.

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-B3-Q1

Grade 6 • Expressions & Equations • 6.EE.A.2 • Set 68 • BACKWARD 3 • Q1/5

Evaluate $x^2 + y^2$ when $x = 4$, $y = -3$.

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-B3-Q2

Grade 6 • Expressions & Equations • 6.EE.A.2 • Set 68 • BACKWARD 3 • Q2/5

$(x + 1)^2$ when $x = -4$ is

A. -9

B. 9

C. 25

D. -3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-B3-Q3

Grade 6 • Expressions & Equations • 6.EE.A.2 • Set 68 • BACKWARD 3 • Q3/5

True or false?

$(-5 - 1)^2 = 36$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-B3-Q4

Grade 6 • Expressions & Equations • 6.EE.A.2 • Set 68 • BACKWARD 3 • Q4/5

True or false?

$(2 + 1)^2 = 2^2 + 1^2$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-B3-Q5

Grade 6 • Expressions & Equations • 6.EE.A.2 • Set 68 • BACKWARD 3 • Q5/5

SET 68 • HSG-GPE.B.4

FORWARD 1

Coordinate proofs about circles with variables • High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4

A(−r, 0) and B(r, 0) are endpoints of a diameter of the circle $x^2 + y^2 = r^2$, and P(x, y) is any other point on the circle. Prove that ∠APB is a right angle.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-F1-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 68 • FORWARD 1 • Q1/5

Show that if (a, b) is on $x^2 + y^2 = r^2$, then (b, a) and (−a, b) are too.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-F1-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 68 • FORWARD 1 • Q2/5

True or false?

A coordinate proof with variables covers every case at once.

A. True

B. False

To prove four points lie on one circle, show they are

A. all the same distance from one point

B. the same distance from the origin only

C. vertices of a rectangle only

D. all on one line

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-F1-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 68 • FORWARD 1 • Q4/5

Prove or disprove: A(1, 0), B(0, 1), C(−1, 0), D(0, −2) lie on one circle centered at the origin.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-F1-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 68 • FORWARD 1 • Q5/5

SET 68 • HSG-GPE.B.4

FORWARD 2

Find the tangent line to a circle at a point • High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5

Find the equation of the line tangent to $x^2 + y^2 = 25$ at $(3, 4)$.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-F2-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 68 • FORWARD 2 • Q1/5

Find the tangent to $(x - 1)^2 + (y - 1)^2 = 25$ at $(4, 5)$.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-F2-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 68 • FORWARD 2 • Q2/5

True or false?

The tangent at P is parallel to the radius drawn to P.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-F2-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 68 • FORWARD 2 • Q3/5

The radius to (0, 5) on $x^2 + y^2 = 25$ is vertical, so the tangent there is

A. $x = 0$

B. $x = 5$

C. $y = x + 5$

D. $y = 5$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-F2-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 68 • FORWARD 2 • Q4/5

Explain how the slope of the tangent at (a, b) on $x^2 + y^2 = r^2$ is found.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-F2-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 68 • FORWARD 2 • Q5/5

Prove the slope criteria for parallel and perpendicular lines

MAIN	Geometry main question	**HSG-GPE.B.5**
BACKWARD 1	Similar slope triangles give the same slope	**8.EE.B.6**
BACKWARD 2	Rotate a point 90° about the origin	**HSG-CO.A.5**
BACKWARD 3	Negative reciprocals	**7.NS.A.2**
FORWARD 1	Use the criteria to classify pairs of lines	**HSG-GPE.B.5**
FORWARD 2	Use the criteria inside a general coordinate proof	**HSG-GPE.B.4**

SET 69 • HSG-GPE.B.5

MAIN

Geometry main question • High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5

Prove that two distinct nonvertical lines with the same slope m are parallel.

Work space

Conclusion: _____

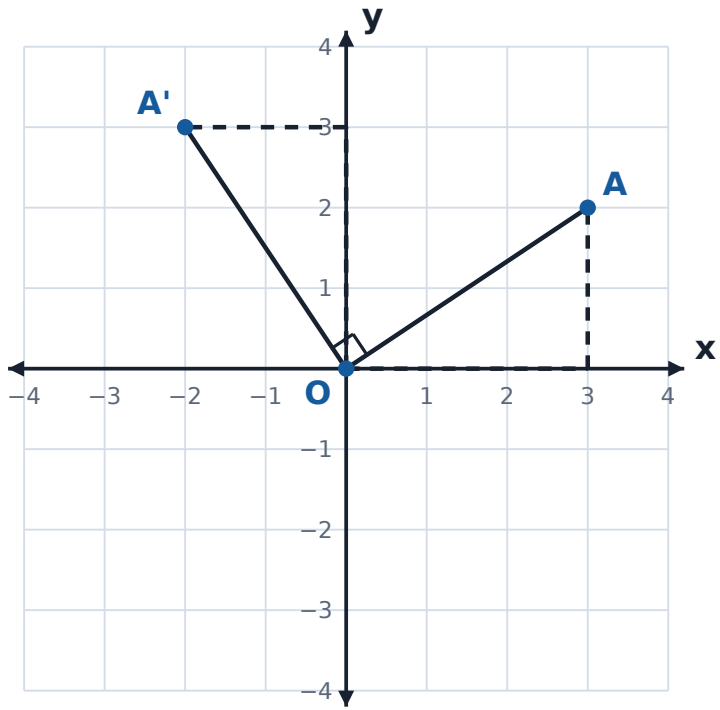
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-M-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 69 • MAIN • Q1/5

Line OA passes through O(0, 0) and A(a, b) with a, b ≠ 0. Rotating A 90° counterclockwise about O gives A'(−b, a). Use this to prove that perpendicular lines have slopes whose product is −1. The diagram shows the case A(3, 2).

Work space



Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-M-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 69 • MAIN • Q2/5

Explain why a 90° rotation of the slope triangle with run 3 and rise 2 gives a slope triangle with run −2 and rise 3.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-M-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 69 • MAIN • Q3/5

Why do two nonvertical lines with equal slopes and different y-intercepts never meet?

A. Their slopes are negative reciprocals

B. Their intercepts are equal

C. Setting the equations equal leads to a false statement

D. They meet at the origin

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-M-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 69 • MAIN • Q4/5

True or false?

Lines with slopes 2 and −2 are perpendicular.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-M-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 69 • MAIN • Q5/5

SET 69 • HSG-GPE.B.5

BACKWARD 1

Similar slope triangles give the same slope • Grade 8 • Expressions & Equations • 8.EE.B.6

True or false?

Any two slope triangles on the same line are similar.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-B1-Q1

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 69 • BACKWARD 1 • Q1/5

A line passes through (0, 0) and (2, 3). Find the rise for a run of 4.

Rise: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-B1-Q2

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 69 • BACKWARD 1 • Q2/5

Slope triangles with run 2, rise 3 and run 6, rise 9 give slopes

A. 6 and 18

B. 3/2 and 9/6 that differ

C. both 3/2

D. 2/3 and 6/9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-B1-Q3

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 69 • BACKWARD 1 • Q3/5

Explain why the slope of a line is the same between any two of its points.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-B1-Q4

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 69 • BACKWARD 1 • Q4/5

True or false?

The slope between two points on a line depends on which points are chosen.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-B1-Q5

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 69 • BACKWARD 1 • Q5/5

SET 69 • HSG-GPE.B.5

BACKWARD 2

Rotate a point 90° about the origin • High School — Geometry • Geometry: Congruence • HSG-CO.A.5

Rotate (3, 2) 90° counterclockwise about the origin.

Image: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-B2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 69 • BACKWARD 2 • Q1/5

Rotate (1, 4) 90° counterclockwise about the origin.

Image: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-B2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 69 • BACKWARD 2 • Q2/5

The rule for a 90° counterclockwise rotation about the origin is

A. $(x, y) \to (-y, x)$

B. $(x, y) \to (-x, -y)$

C. $(x, y) \to (y, x)$

D. $(x, y) \to (y, -x)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-B2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 69 • BACKWARD 2 • Q3/5

True or false?

A 90° rotation maps a line to a perpendicular line.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-B2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 69 • BACKWARD 2 • Q4/5

True or false?

A 90° rotation maps (2, 0) to (0, −2).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-B2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 69 • BACKWARD 2 • Q5/5

SET 69 • HSG-GPE.B.5

BACKWARD 3

Negative reciprocals • Grade 7 • The Number System • 7.NS.A.2

Negative reciprocal of 2/3?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-B3-Q1

Grade 7 • The Number System • 7.NS.A.2 • Set 69 • BACKWARD 3 • Q1/5

Multiply $(b/a)(-a/b)$.

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-B3-Q2

Grade 7 • The Number System • 7.NS.A.2 • Set 69 • BACKWARD 3 • Q2/5

Negative reciprocal of -4?

A. $\frac{1}{4}$

B. $-\frac{1}{4}$

C. 4

D. -4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-B3-Q3

Grade 7 • The Number System • 7.NS.A.2 • Set 69 • BACKWARD 3 • Q3/5

True or false?

A number times its negative reciprocal is -1.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-B3-Q4

Grade 7 • The Number System • 7.NS.A.2 • Set 69 • BACKWARD 3 • Q4/5

True or false?

The negative reciprocal of 1 is 1.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-B3-Q5

Grade 7 • The Number System • 7.NS.A.2 • Set 69 • BACKWARD 3 • Q5/5

SET 69 • HSG-GPE.B.5

FORWARD 1

Use the criteria to classify pairs of lines • High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5

Are $y = 3x - 2$ and $6x - 2y = 7$ parallel, perpendicular, or neither?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-F1-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 69 • FORWARD 1 • Q1/5

Are $2x + 5y = 1$ and $5x - 2y = 4$ parallel, perpendicular, or neither?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-F1-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 69 • FORWARD 1 • Q2/5

True or false?

A horizontal line and a vertical line are perpendicular even though the slope product rule cannot be used.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-F1-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 69 • FORWARD 1 • Q3/5

$y = \frac{1}{2}x + 1$ and $y = 2x - 1$ are

A. neither

B. the same line

C. perpendicular

D. parallel

Find k so that $y = kx + 1$ is perpendicular to $y = -\frac{1}{3}x$.

$k =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-F1-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 69 • FORWARD 1 • Q5/5

SET 69 • HSG-GPE.B.5

FORWARD 2

Use the criteria inside a general coordinate proof • High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4

Rhombus OABC has O(0, 0), A(a, 0), B(a + b, c), C(b, c), with $b^2 + c^2 = a^2$. Prove that its diagonals are perpendicular.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-F2-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 69 • FORWARD 2 • Q1/5

A coordinate proof places rhombus OABC at O(0, 0), A(a, 0), B(a + b, c), C(b, c). Why must the proof assume $b^2 + c^2 = a^2$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-F2-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 69 • FORWARD 2 • Q2/5

True or false?

The slope criterion turns "perpendicular" into an algebra statement.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-F2-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 69 • FORWARD 2 • Q3/5

In a coordinate proof that lines are parallel, you show

A. equal slopes

B. equal intercepts

C. slope product -1

D. equal lengths

Prove or disprove with slopes: A(0, 0), B(4, 2), C(3, 4) form a right angle at B.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-F2-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 69 • FORWARD 2 • Q5/5

Write equations of parallel and perpendicular lines

MAIN	Geometry main question	**HSG-GPE.B.5**
BACKWARD 1	Read the slope of a line	**8.F.B.4**
BACKWARD 2	Rewrite an equation in slope-intercept form	**HSA-CED.A.4**
BACKWARD 3	Write a line from a point and a slope	**HSF-LE.A.2**
BACKWARD 4	Opposite reciprocals	**7.NS.A.2**
FORWARD 1	Write perpendicular bisectors and altitudes	**HSG-GPE.B.5**
FORWARD 2	Find the distance from a point to a line	**HSG-GPE.B.4**

SET 70 • HSG-GPE.B.5

MAIN

Geometry main question • High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5

Write the equation of the line parallel to $y = 3x - 5$ that passes through (2, 4).

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-M-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 70 • MAIN • Q1/5

Write the equation of the line perpendicular to $y = \frac{2}{3}x + 1$ that passes through $(4, -1)$.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-M-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 70 • MAIN • Q2/5

Write the equation of the line parallel to $2x + 5y = 10$ that passes through (5, 3).

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-M-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 70 • MAIN • Q3/5

Write the equation of the line perpendicular to $x = 4$ that passes through $(1, -3)$.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-M-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 70 • MAIN • Q4/5

Which line is perpendicular to $y = -4x + 1$ and passes through the origin?

A. $y = -4x$

B. $y = -\frac{1}{4}x$

C. $y = \frac{1}{4}x$

D. $y = 4x$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-M-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 70 • MAIN • Q5/5

SET 70 • HSG-GPE.B.5

BACKWARD 1

Read the slope of a line • Grade 8 • Functions • 8.F.B.4

Slope of $y = -2x + 7$?

Slope: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-B1-Q1

Grade 8 • Functions • 8.F.B.4 • Set 70 • BACKWARD 1 • Q1/5

Slope of $y = 5$?

Slope: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-B1-Q2

Grade 8 • Functions • 8.F.B.4 • Set 70 • BACKWARD 1 • Q2/5

Slope of $y = \frac{2}{3}x - 4$?

A. 3/2

B. $\frac{2}{3}$

C. -4

D. 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-B1-Q3

Grade 8 • Functions • 8.F.B.4 • Set 70 • BACKWARD 1 • Q3/5

True or false?

In $y = mx + b$, m is the slope.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-B1-Q4

Grade 8 • Functions • 8.F.B.4 • Set 70 • BACKWARD 1 • Q4/5

True or false?

The slope of $x = 3$ is 3.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-B1-Q5

Grade 8 • Functions • 8.F.B.4 • Set 70 • BACKWARD 1 • Q5/5

SET 70 • HSG-GPE.B.5

BACKWARD 2

Rewrite an equation in slope-intercept form • High School — Algebra I (course placement in this bank) • Algebra: Creating Equations • HSA-CED.A.4

Solve $2x + 5y = 10$ for y.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-B2-Q1

High School — Algebra I (course placement in this bank) • Algebra: Creating Equations • HSA-CED.A.4 • Set 70 • BACKWARD 2 • Q1/5

Solve $3x - y = 4$ for y.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-B2-Q2

High School — Algebra I (course placement in this bank) • Algebra: Creating Equations • HSA-CED.A.4 • Set 70 • BACKWARD 2 • Q2/5

$6x + 3y = 9$ in slope-intercept form is

A. $y = 2x + 3$

B. $y = -6x + 9$

C. $y = -2x + 9$

D. $y = -2x + 3$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-B2-Q3

High School — Algebra I (course placement in this bank) • Algebra: Creating Equations • HSA-CED.A.4 • Set 70 • BACKWARD 2 • Q3/5

True or false?

$x - 2y = 6$ has slope ½.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-B2-Q4

High School — Algebra I (course placement in this bank) • Algebra: Creating Equations • HSA-CED.A.4 • Set 70 • BACKWARD 2 • Q4/5

True or false?

$4x + 2y = 8$ has slope 4.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-B2-Q5

High School — Algebra I (course placement in this bank) • Algebra: Creating Equations • HSA-CED.A.4 • Set 70 • BACKWARD 2 • Q5/5

SET 70 • HSG-GPE.B.5

BACKWARD 3

Write a line from a point and a slope • High School — Algebra I (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2

Write the line with slope 3 through (2, 4).

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-B3-Q1

High School — Algebra I (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 70 • BACKWARD 3 • Q1/5

Write the line with slope -1 through $(0, 6)$.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-B3-Q2

High School — Algebra I (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 70 • BACKWARD 3 • Q2/5

Slope 2 through (1, 5): b =

A. 5

B. 7

C. −3

D. 3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S70-B3-Q3**

High School — Algebra I (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 70 • BACKWARD 3 • Q3/5

True or false?

$y - 4 = 3(x - 2)$ is the line with slope 3 through (2, 4).

A. True

B. False

True or false?

A line through (0, 6) has y-intercept 0.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-B3-Q5

High School — Algebra I (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 70 • BACKWARD 3 • Q5/5

SET 70 • HSG-GPE.B.5

BACKWARD 4

Opposite reciprocals • Grade 7 • The Number System • 7.NS.A.2

Negative reciprocal of ⅔?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-B4-Q1

Grade 7 • The Number System • 7.NS.A.2 • Set 70 • BACKWARD 4 • Q1/5

Negative reciprocal of -5?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-B4-Q2

Grade 7 • The Number System • 7.NS.A.2 • Set 70 • BACKWARD 4 • Q2/5

Negative reciprocal of −2/5?

A. 5/2

B. 2/5

C. −2/5

D. −5/2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-B4-Q3

Grade 7 • The Number System • 7.NS.A.2 • Set 70 • BACKWARD 4 • Q3/5

True or false?

The negative reciprocal of ¼ is −4.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-B4-Q4

Grade 7 • The Number System • 7.NS.A.2 • Set 70 • BACKWARD 4 • Q4/5

True or false?

The negative reciprocal of 3 is −3.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-B4-Q5

Grade 7 • The Number System • 7.NS.A.2 • Set 70 • BACKWARD 4 • Q5/5

SET 70 • HSG-GPE.B.5

FORWARD 1

Write perpendicular bisectors and altitudes • High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5

Write the equation of the perpendicular bisector of the segment from (2, 1) to (6, 5).

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-F1-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 70 • FORWARD 1 • Q1/5

Triangle ABC has A(0, 0), B(6, 0), C(2, 4). Write the equation of the altitude from A to BC.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-F1-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 70 • FORWARD 1 • Q2/5

True or false?

A perpendicular bisector passes through the midpoint of the segment.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-F1-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 70 • FORWARD 1 • Q3/5

The perpendicular bisector of the segment from (0, 0) to (0, 8) is

A. $x = 0$

B. $y = 4$

C. $x = 4$

D. $y = 8$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-F1-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 70 • FORWARD 1 • Q4/5

Write the perpendicular bisector of the segment from $(-3, 2)$ to $(5, 2)$.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-F1-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 70 • FORWARD 1 • Q5/5

SET 70 • HSG-GPE.B.5

FORWARD 2

Find the distance from a point to a line • High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4

Find the distance from P(5, 5) to the line $y = -x + 2$.

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-F2-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 70 • FORWARD 2 • Q1/5

Find the distance from (0, 0) to the line y = 4.

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-F2-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 70 • FORWARD 2 • Q2/5

True or false?

The distance from a point to a line can be measured along any segment from the point to the line.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-F2-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 70 • FORWARD 2 • Q3/5

The foot of the perpendicular from (3, 0) to y = x is

A. (0, 3)

B. (0, 0)

C. (3, 3)

D. (1.5, 1.5)

Find the distance from (6, 1) to the line $y = 2x - 1$.

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-F2-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 70 • FORWARD 2 • Q5/5

Solve geometric problems with the slope criteria

MAIN	Geometry main question	**HSG-GPE.B.5**
BACKWARD 1	Slope from two points	**8.F.B.4**
BACKWARD 2	Solve a linear equation for an unknown	**HSA-REI.B.3**
BACKWARD 3	Equations of parallel and perpendicular lines	**HSG-GPE.B.5**
BACKWARD 4	Opposite sides of a parallelogram are parallel	**HSG-CO.C.11**
FORWARD 1	Find the orthocenter of a triangle	**HSG-GPE.B.5**
FORWARD 2	Shortest path to a straight line in a design	**HSG-MG.A.3**

SET 71 • HSG-GPE.B.5

MAIN

Geometry main question • High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5

The line through (1, 2) and (4, k) is parallel to $y = 2x + 7$. Find k.

$k =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-M-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 71 • MAIN • Q1/5

ABCD is a parallelogram with A(0, 0), B(5, 1), C(7, 4). Find D.

D = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-M-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 71 • MAIN • Q2/5

The segment from (2, 1) to (5, k) is perpendicular to the line y = 3x. Find k.

k = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-M-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 71 • MAIN • Q3/5

Triangle ABC has A(0, 0), B(8, 4), C(4, 6). Find the foot of the altitude from C to AB.

Foot: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-M-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 71 • MAIN • Q4/5

Triangle ABC has A(1, 1), B(4, 2), C(3, 5). At which vertex is the right angle?

A. C

B. B

C. A

D. There is no right angle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-M-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 71 • MAIN • Q5/5

SET 71 • HSG-GPE.B.5

BACKWARD 1

Slope from two points • Grade 8 • Functions • 8.F.B.4

Slope through (1, 2) and (4, 8)?

Slope: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-B1-Q1

Grade 8 • Functions • 8.F.B.4 • Set 71 • BACKWARD 1 • Q1/5

Slope through (4, 2) and (3, 5)?

Slope: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-B1-Q2

Grade 8 • Functions • 8.F.B.4 • Set 71 • BACKWARD 1 • Q2/5

Slope through (0, 0) and (8, 4)?

A. ½

B. 8

C. 4

D. 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-B1-Q3

Grade 8 • Functions • 8.F.B.4 • Set 71 • BACKWARD 1 • Q3/5

True or false?

The slope from A to B equals the slope from B to A.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-B1-Q4

Grade 8 • Functions • 8.F.B.4 • Set 71 • BACKWARD 1 • Q4/5

True or false?

The slope through (1, 1) and (1, 5) is 0.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-B1-Q5

Grade 8 • Functions • 8.F.B.4 • Set 71 • BACKWARD 1 • Q5/5

SET 71 • HSG-GPE.B.5

BACKWARD 2

Solve a linear equation for an unknown • High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3

Solve $(k - 2)/3 = 2$.

$k =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-B2-Q1

High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 71 • BACKWARD 2 • Q1/5

Solve ½x = −2x + 14.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-B2-Q2

High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 71 • BACKWARD 2 • Q2/5

Solve $(k - 1)/3 = -\frac{1}{3}$.

A. -2

B. 1

C. 2

D. 0

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S71-B2-Q3**

High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 71 • BACKWARD 2 • Q3/5

True or false?

Multiplying both sides by 3 is a valid first step in $(k - 2)/3 = 2$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-B2-Q4

High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 71 • BACKWARD 2 • Q4/5

True or false?

The solution of $2x + 1 = 7$ is 4.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-B2-Q5

High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 71 • BACKWARD 2 • Q5/5

SET 71 • HSG-GPE.B.5

BACKWARD 3

Equations of parallel and perpendicular lines • High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5

Line perpendicular to y = ½x through (4, 6)?

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-B3-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 71 • BACKWARD 3 • Q1/5

Line parallel to $y = -x$ through $(0, 3)$?

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-B3-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 71 • BACKWARD 3 • Q2/5

Perpendicular slope to ⅓?

A. 3

B. −3

C. −⅓

D. ⅓

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-B3-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 71 • BACKWARD 3 • Q3/5

True or false?

An altitude is perpendicular to the side it meets.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-B3-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 71 • BACKWARD 3 • Q4/5

True or false?

An altitude passes through the midpoint of the opposite side.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-B3-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 71 • BACKWARD 3 • Q5/5

BACKWARD 4

Opposite sides of a parallelogram are parallel • High School — Geometry • Geometry: Congruence • HSG-CO.C.11

True or false?

In a parallelogram, both pairs of opposite sides are parallel and congruent.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-B4-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 71 • BACKWARD 4 • Q1/5

A parallelogram has A(0, 0), B(3, 0), C(4, 2). Find D (in order ABCD).

D = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-B4-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 71 • BACKWARD 4 • Q2/5

In parallelogram ABCD, side AB is parallel to

A. AC

B. BC

C. AD

D. DC

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-B4-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 71 • BACKWARD 4 • Q3/5

Going from B to C is "right 2, up 3". Going from A to D is

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-B4-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 71 • BACKWARD 4 • Q4/5

True or false?

A parallelogram's adjacent sides are always perpendicular.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-B4-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 71 • BACKWARD 4 • Q5/5

SET 71 • HSG-GPE.B.5

FORWARD 1

Find the orthocenter of a triangle • High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5

Triangle ABC has A(0, 0), B(6, 0), C(2, 4). Find the orthocenter (the intersection of the altitudes).

Orthocenter: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-F1-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 71 • FORWARD 1 • Q1/5

Triangle with A(0, 0), B(4, 0), C(0, 3). Where is the orthocenter?

Orthocenter: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-F1-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 71 • FORWARD 1 • Q2/5

True or false?

The orthocenter of a right triangle is at the right-angle vertex.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-F1-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 71 • FORWARD 1 • Q3/5

To find the orthocenter, you need at least

A. three medians

B. the circumcenter

C. one altitude

D. two altitudes

Check that the third altitude of the triangle A(0, 0), B(6, 0), C(2, 4) also passes through (2, 2).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-F1-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 71 • FORWARD 1 • Q5/5

SET 71 • HSG-GPE.B.5

FORWARD 2

Shortest path to a straight line in a design • High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3

A well is at W(2, 9). A straight canal runs along $y = 2x - 1$ (units: 100 m). Where should a pipe from the well meet the canal so the pipe is as short as possible, and how long is it?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-F2-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 71 • FORWARD 2 • Q1/5

A pipe will connect a well to a straight canal. Why is the shortest possible pipe perpendicular to the canal?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-F2-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 71 • FORWARD 2 • Q2/5

True or false?

A pipe meeting the canal at a 60° angle is shorter than one meeting it at 90°.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-F2-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 71 • FORWARD 2 • Q3/5

A path from (0, 5) to the road $y = 0$ is shortest at

A. (0, 0)

B. (5, 0)

C. (−5, 0)

D. (0, 5)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-F2-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 71 • FORWARD 2 • Q4/5

A sensor at (3, 4) must connect to the line y = x. Where should the wire meet the line?

Point: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-F2-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 71 • FORWARD 2 • Q5/5

Partition a directed segment in a given ratio

MAIN	Geometry main question	**HSG-GPE.B.6**
BACKWARD 1	A ratio as a fraction of the whole	**6.RP.A.1**
BACKWARD 2	Horizontal and vertical change	**6.NS.C.8**
BACKWARD 3	Find a fraction of a number	**6.NS.A.1**
FORWARD 1	Work backward: find an endpoint or the ratio	**HSG-GPE.B.6**
FORWARD 2	The centroid as a partition point	**HSG-GPE.B.4**

SET 72 • HSG-GPE.B.6

MAIN

Geometry main question • High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.6

Find the point P on the directed segment from A(1, 2) to B(13, 8) that partitions it in the ratio 1 : 2.

P = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S72-M-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.6 • Set 72 • MAIN • Q1/5

Find the point P on the directed segment from A(−4, 6) to B(6, 1) that partitions it in the ratio 3 : 2.

P = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S72-M-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.6 • Set 72 • MAIN • Q2/5

Find the point P on the directed segment from A(2, −3) to B(−6, 9) that partitions it in the ratio 1 : 3.

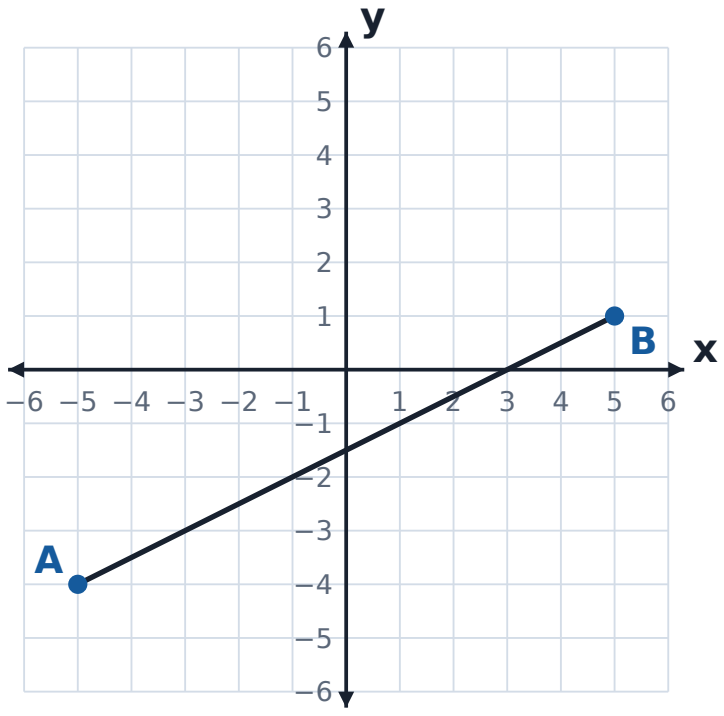
P = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S72-M-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.6 • Set 72 • MAIN • Q3/5

Find the point P on the directed segment from A to B that partitions it in the ratio 2 : 3.



P = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S72-M-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.6 • Set 72 • MAIN • Q4/5

Point P partitions the directed segment from B(10, 5) to A(0, 0) in the ratio 1 : 4. P is

A. (8, 4)

B. (2, 1)

C. (5, 2.5)

D. (4, 8)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S72-M-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.6 • Set 72 • MAIN • Q5/5

SET 72 • HSG-GPE.B.6

BACKWARD 1

A ratio as a fraction of the whole • Grade 6 • Ratios & Proportional Relationships • 6.RP.A.1

A ratio of 2 : 3 splits a whole into how many equal parts?

Parts: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S72-B1-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.1 • Set 72 • BACKWARD 1 • Q1/5

In a 1 : 2 partition, the first part is what fraction of the whole?

Fraction: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S72-B1-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.1 • Set 72 • BACKWARD 1 • Q2/5

In a 3 : 2 partition, the first part is

A. 3/2 of the whole

B. 2/3 of the whole

C. 3/5 of the whole

D. 2/5 of the whole

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S72-B1-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.1 • Set 72 • BACKWARD 1 • Q3/5

True or false?

A ratio of 1 : 3 means the point is ¼ of the way along.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S72-B1-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.1 • Set 72 • BACKWARD 1 • Q4/5

True or false?

A ratio of 1 : 2 means the point is halfway.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S72-B1-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.1 • Set 72 • BACKWARD 1 • Q5/5

SET 72 • HSG-GPE.B.6

BACKWARD 2

Horizontal and vertical change • Grade 6 • The Number System • 6.NS.C.8

From (1, 2) to (13, 8), what is the change in x?

Change: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S72-B2-Q1

Grade 6 • The Number System • 6.NS.C.8 • Set 72 • BACKWARD 2 • Q1/5

From (−4, 6) to (6, 1), what is the change in y?

Change: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S72-B2-Q2

Grade 6 • The Number System • 6.NS.C.8 • Set 72 • BACKWARD 2 • Q2/5

From $(2, -3)$ to $(-6, 9)$, the change in x is

A. -8

B. -4

C. 4

D. 8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S72-B2-Q3

Grade 6 • The Number System • 6.NS.C.8 • Set 72 • BACKWARD 2 • Q3/5

True or false?

Change in x is the ending x minus the starting x.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S72-B2-Q4

Grade 6 • The Number System • 6.NS.C.8 • Set 72 • BACKWARD 2 • Q4/5

True or false?

The change in y from (0, 5) to (0, 1) is 4.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S72-B2-Q5

Grade 6 • The Number System • 6.NS.C.8 • Set 72 • BACKWARD 2 • Q5/5

SET 72 • HSG-GPE.B.6

BACKWARD 3

Find a fraction of a number • Grade 6 • The Number System • 6.NS.A.1

Find ⅓ of 12.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S72-B3-Q1

Grade 6 • The Number System • 6.NS.A.1 • Set 72 • BACKWARD 3 • Q1/5

Find 3/5 of −5.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S72-B3-Q2

Grade 6 • The Number System • 6.NS.A.1 • Set 72 • BACKWARD 3 • Q2/5

2/5 of 10 is

A. 5

B. 25

C. 4

D. 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S72-B3-Q3

Grade 6 • The Number System • 6.NS.A.1 • Set 72 • BACKWARD 3 • Q3/5

True or false?

¼ of −8 is −2.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S72-B3-Q4

Grade 6 • The Number System • 6.NS.A.1 • Set 72 • BACKWARD 3 • Q4/5

True or false?

⅔ of 9 is 3.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S72-B3-Q5

Grade 6 • The Number System • 6.NS.A.1 • Set 72 • BACKWARD 3 • Q5/5

SET 72 • HSG-GPE.B.6

FORWARD 1

Work backward: find an endpoint or the ratio • High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.6

P(3, 4) partitions the directed segment from A(1, 2) to B in the ratio 1 : 2. Find B.

B = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S72-F1-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.6 • Set 72 • FORWARD 1 • Q1/5

P(4, 3) is on the segment from A(0, 1) to B(10, 6). In what ratio does P partition AB?

Ratio: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S72-F1-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.6 • Set 72 • FORWARD 1 • Q2/5

True or false?

The midpoint partitions a segment in the ratio $1 : 1$.

A. True

B. False

P partitions AB in 3 : 1. Then P is closer to

A. B

B. neither

C. A

D. the origin

M(2, 5) is the midpoint of AB with A(−1, 1). Find B.

B = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S72-F1-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.6 • Set 72 • FORWARD 1 • Q5/5

SET 72 • HSG-GPE.B.6

FORWARD 2

The centroid as a partition point • High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4

Triangle with A(0, 0), B(6, 0), C(3, 9). Find the point ⅔ of the way from A to the midpoint of BC.

Point: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S72-F2-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 72 • FORWARD 2 • Q1/5

Show that the centroid (the point 2 : 1 along a median from the vertex) of the triangle with vertices (x_1, y_1), (x_2, y_2), (x_3, y_3) is the average of the vertices.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S72-F2-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 72 • FORWARD 2 • Q2/5

True or false?

The centroid of the triangle (0, 0), (6, 0), (3, 9) is (3, 4.5).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S72-F2-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 72 • FORWARD 2 • Q3/5

The centroid of (1, 2), (4, 8), (7, −1) is

A. (4, 3)

B. (4, 9)

C. (3, 3)

D. (12, 9)

Find the centroid of the triangle with vertices $(-2, 5)$, $(4, 1)$, $(7, 0)$.

Centroid: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S72-F2-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 72 • FORWARD 2 • Q5/5

SET 73 • HSG-GPE.B.7 • EXPRESSING GEOMETRIC PROPERTIES WITH EQUATIONS

Compute perimeters of polygons with coordinates

MAIN	Geometry main question	**HSG-GPE.B.7**
BACKWARD 1	Distance between two points	**8.G.B.8**
BACKWARD 2	Lengths of horizontal and vertical sides	**6.G.A.3**
BACKWARD 3	Simplify square roots	**HSN-RN.A.2**
FORWARD 1	Find a missing coordinate from a perimeter	**HSG-GPE.B.7**
FORWARD 2	Fencing and cost from a coordinate plan	**HSG-MG.A.3**

SET 73 • HSG-GPE.B.7

MAIN

Geometry main question • High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7

Find the perimeter of the triangle with vertices (0, 0), (3, 4), and (3, 0).

Perimeter: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S73-M-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 73 • MAIN • Q1/5

Find the perimeter of the quadrilateral with vertices A(1, 1), B(4, 5), C(10, −3), D(7, −7).

Perimeter: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S73-M-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 73 • MAIN • Q2/5

Find the perimeter of the triangle with vertices (0, 0), (4, 0), (2, 4). Give the exact value and a decimal to the nearest tenth.

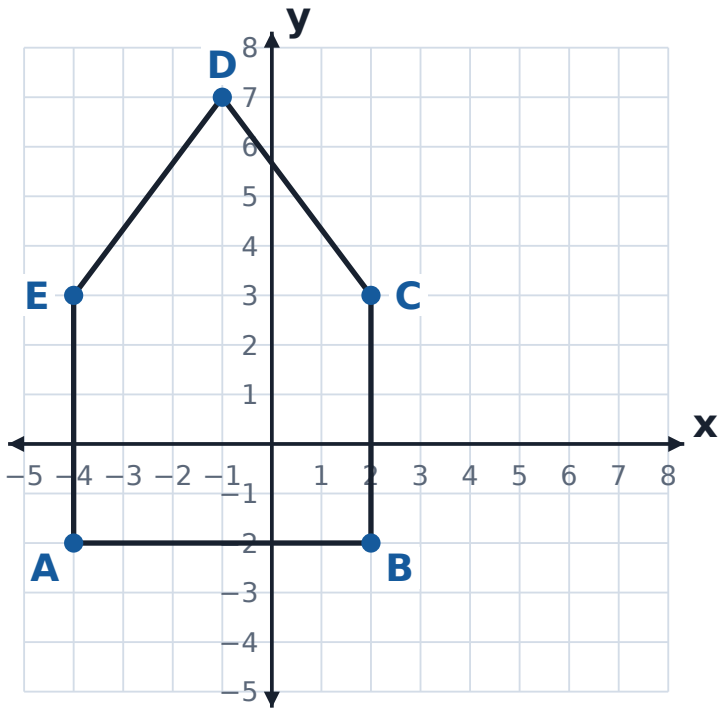
Perimeter: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S73-M-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 73 • MAIN • Q3/5

Find the perimeter of pentagon ABCDE.



Perimeter: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S73-M-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 73 • MAIN • Q4/5

A square has vertices (0, 0), (3, 1), (2, 4), (−1, 3). Its perimeter is

A. $4\sqrt{10}$

B. 10

C. 40

D. 16

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S73-M-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 73 • MAIN • Q5/5

SET 73 • HSG-GPE.B.7

BACKWARD 1

Distance between two points • Grade 8 • Geometry • 8.G.B.8

Distance from (2, 3) to (−1, 7)?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S73-B1-Q1

Grade 8 • Geometry • 8.G.B.8 • Set 73 • BACKWARD 1 • Q1/5

Distance from (4, 5) to (10, −3)?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S73-B1-Q2

Grade 8 • Geometry • 8.G.B.8 • Set 73 • BACKWARD 1 • Q2/5

Distance from (0, 0) to (2, 4)?

A. 20

B. $2\sqrt{5}$

C. 6

D. $\sqrt{6}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S73-B1-Q3

Grade 8 • Geometry • 8.G.B.8 • Set 73 • BACKWARD 1 • Q3/5

True or false?

The distance from (0, 0) to (3, 1) is √10.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S73-B1-Q4

Grade 8 • Geometry • 8.G.B.8 • Set 73 • BACKWARD 1 • Q4/5

True or false?

Distances can be negative.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S73-B1-Q5

Grade 8 • Geometry • 8.G.B.8 • Set 73 • BACKWARD 1 • Q5/5

SET 73 • HSG-GPE.B.7

BACKWARD 2

Lengths of horizontal and vertical sides • Grade 6 • Geometry • 6.G.A.3

Length of the side from $(-4, -2)$ to $(2, -2)$?

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S73-B2-Q1

Grade 6 • Geometry • 6.G.A.3 • Set 73 • BACKWARD 2 • Q1/5

Length of the side from (2, −2) to (2, 3)?

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S73-B2-Q2

Grade 6 • Geometry • 6.G.A.3 • Set 73 • BACKWARD 2 • Q2/5

Length from $(-3, 4)$ to $(-3, -1)$?

A. 7

B. 3

C. 5

D. -5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S73-B2-Q3

Grade 6 • Geometry • 6.G.A.3 • Set 73 • BACKWARD 2 • Q3/5

True or false?

A horizontal side's length is the difference of the x-coordinates.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S73-B2-Q4

Grade 6 • Geometry • 6.G.A.3 • Set 73 • BACKWARD 2 • Q4/5

True or false?

The side from (1, 1) to (1, 9) has length 10.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S73-B2-Q5

Grade 6 • Geometry • 6.G.A.3 • Set 73 • BACKWARD 2 • Q5/5

SET 73 • HSG-GPE.B.7

BACKWARD 3

Simplify square roots • High School — Algebra I (course placement in this bank) • Number & Quantity: The Real Number System • HSN-RN.A.2

Simplify $\sqrt{20}$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S73-B3-Q1

High School — Algebra I (course placement in this bank) • Number & Quantity: The Real Number System • HSN-RN.A.2 • Set 73 • BACKWARD 3 • Q1/5

Simplify $\sqrt{50}$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S73-B3-Q2

High School — Algebra I (course placement in this bank) • Number & Quantity: The Real Number System • HSN-RN.A.2 • Set 73 • BACKWARD 3 • Q2/5

$2\sqrt{5} + 2\sqrt{5} =$

A. $2\sqrt{10}$

B. $8\sqrt{5}$

C. $4\sqrt{10}$

D. $4\sqrt{5}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S73-B3-Q3**

High School — Algebra I (course placement in this bank) • Number & Quantity: The Real Number System • HSN-RN.A.2 • Set 73 • BACKWARD 3 • Q3/5

True or false?

$\sqrt{10} \approx 3.16$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S73-B3-Q4

High School — Algebra I (course placement in this bank) • Number & Quantity: The Real Number System • HSN-RN.A.2 • Set 73 • BACKWARD 3 • Q4/5

True or false?

$\sqrt{9} + \sqrt{16} = \sqrt{25}$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S73-B3-Q5

High School — Algebra I (course placement in this bank) • Number & Quantity: The Real Number System • HSN-RN.A.2 • Set 73 • BACKWARD 3 • Q5/5

SET 73 • HSG-GPE.B.7

FORWARD 1

Find a missing coordinate from a perimeter • High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7

A rectangle has vertices (0, 0), (k, 0), (k, 3), (0, 3) with $k > 0$ and perimeter 22. Find k.

$k =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S73-F1-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 73 • FORWARD 1 • Q1/5

Triangle with vertices (0, 0), (6, 0), (3, h), h > 0, has perimeter 16. Find h.

h = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S73-F1-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 73 • FORWARD 1 • Q2/5

True or false?

Doubling every coordinate of a polygon multiplies its perimeter by 4.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S73-F1-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 73 • FORWARD 1 • Q3/5

A square with one side from (0, 0) to (4, 3) has perimeter

A. 28

B. 25

C. 14

D. 20

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S73-F1-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 73 • FORWARD 1 • Q4/5

Is the triangle (0, 0), (6, 0), (3, 4) isosceles? Justify with lengths.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S73-F1-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 73 • FORWARD 1 • Q5/5

SET 73 • HSG-GPE.B.7

FORWARD 2

Fencing and cost from a coordinate plan • High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3

A garden plan has corners (0, 0), (8, 0), (8, 6), (0, 6), in meters. Fencing costs $12 per meter. Find the cost to fence it.

Cost: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S73-F2-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 73 • FORWARD 2 • Q1/5

A triangular lot has corners (0, 0), (30, 0), (30, 40) in feet. Edging costs $2.50 per foot. Find the total cost.

Cost: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S73-F2-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 73 • FORWARD 2 • Q2/5

True or false?

On a map with 1 unit = 5 m, an area of 12 square units is 60 m².

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S73-F2-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 73 • FORWARD 2 • Q3/5

A path has corners (0, 0), (3, 4), (9, 12) (units: 10 m). Its length is

A. 150 m

B. 210 m

C. 15 m

D. 21 m

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S73-F2-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 73 • FORWARD 2 • Q4/5

A run around a lot with corners (0, 0), (5, 12), (5, 0) in units of 100 m: how many meters?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S73-F2-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 73 • FORWARD 2 • Q5/5

Compute areas of triangles and rectangles with coordinates

MAIN	Geometry main question	**HSG-GPE.B.7**
BACKWARD 1	Area of a triangle from base and height	**6.G.A.1**
BACKWARD 2	Lengths on the coordinate plane	**6.G.A.3**
BACKWARD 3	Distance between two points	**8.G.B.8**
FORWARD 1	Find a missing vertex from an area	**HSG-GPE.B.7**
FORWARD 2	Use area from coordinates with density	**HSG-MG.A.2**

SET 74 • HSG-GPE.B.7

MAIN

Geometry main question • High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7

Find the area of the triangle with vertices (0, 0), (8, 0), and (3, 5).

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S74-M-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 74 • MAIN • Q1/5

Rectangle ABCD has A(1, 1), B(5, 3), C(4, 5), D(0, 3). Find its area.

Area: _____

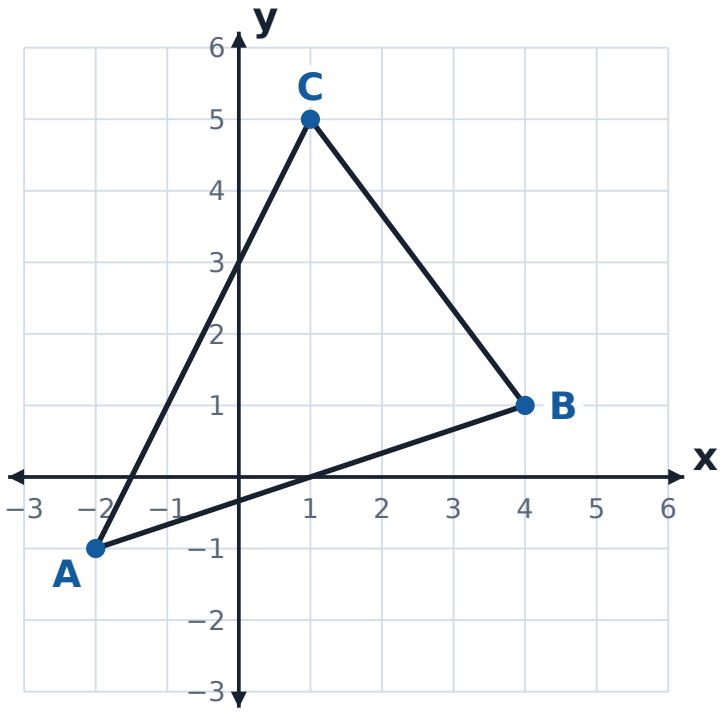
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S74-M-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 74 • MAIN • Q2/5

Find the area of triangle ABC.

Area: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S74-M-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 74 • MAIN • Q3/5

Find the area of the triangle with vertices A(0, 0), B(6, 2), C(2, 6).

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S74-M-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 74 • MAIN • Q4/5

A triangle has vertices (0, 0), (10, 0), and (x, 4). What is its area?

A. 20, for any x

B. It depends on x

C. 40

D. 20 only if x = 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S74-M-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 74 • MAIN • Q5/5

SET 74 • HSG-GPE.B.7

BACKWARD 1

Area of a triangle from base and height • Grade 6 • Geometry • 6.G.A.1

Area of a triangle with base 8 and height 5?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S74-B1-Q1

Grade 6 • Geometry • 6.G.A.1 • Set 74 • BACKWARD 1 • Q1/5

Area of a right triangle with legs 6 and 2?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S74-B1-Q2

Grade 6 • Geometry • 6.G.A.1 • Set 74 • BACKWARD 1 • Q2/5

A =

A. ½bh

B. b + h

C. 2bh

D. bh

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S74-B1-Q3

Grade 6 • Geometry • 6.G.A.1 • Set 74 • BACKWARD 1 • Q3/5

True or false?

The height of a triangle is perpendicular to the base.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S74-B1-Q4

Grade 6 • Geometry • 6.G.A.1 • Set 74 • BACKWARD 1 • Q4/5

True or false?

The height must be one of the sides.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S74-B1-Q5

Grade 6 • Geometry • 6.G.A.1 • Set 74 • BACKWARD 1 • Q5/5

SET 74 • HSG-GPE.B.7

BACKWARD 2

Lengths on the coordinate plane • Grade 6 • Geometry • 6.G.A.3

Length from $(-2, -1)$ to $(4, -1)$?

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S74-B2-Q1

Grade 6 • Geometry • 6.G.A.3 • Set 74 • BACKWARD 2 • Q1/5

Height of the point (3, 5) above the x-axis?

Height: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S74-B2-Q2

Grade 6 • Geometry • 6.G.A.3 • Set 74 • BACKWARD 2 • Q2/5

Width of the rectangle enclosing x from −2 to 4?

A. 2

B. 8

C. 4

D. 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S74-B2-Q3

Grade 6 • Geometry • 6.G.A.3 • Set 74 • BACKWARD 2 • Q3/5

True or false?

A point with y-coordinate -3 is 3 units from the x-axis.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S74-B2-Q4

Grade 6 • Geometry • 6.G.A.3 • Set 74 • BACKWARD 2 • Q4/5

True or false?

The length from (0, 2) to (0, 9) is 11.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S74-B2-Q5

Grade 6 • Geometry • 6.G.A.3 • Set 74 • BACKWARD 2 • Q5/5

SET 74 • HSG-GPE.B.7

BACKWARD 3

Distance between two points • Grade 8 • Geometry • 8.G.B.8

Distance from (1, 1) to (5, 3)?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S74-B3-Q1

Grade 8 • Geometry • 8.G.B.8 • Set 74 • BACKWARD 3 • Q1/5

Distance from (5, 3) to (4, 5)?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S74-B3-Q2

Grade 8 • Geometry • 8.G.B.8 • Set 74 • BACKWARD 3 • Q2/5

$\sqrt{20} \cdot \sqrt{5} =$

A. 10

B. 25

C. $\sqrt{25}$

D. 100

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S74-B3-Q3

Grade 8 • Geometry • 8.G.B.8 • Set 74 • BACKWARD 3 • Q3/5

True or false?

Slanted rectangle sides need the distance formula.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S74-B3-Q4

Grade 8 • Geometry • 8.G.B.8 • Set 74 • BACKWARD 3 • Q4/5

True or false?

$\sqrt{20} \cdot \sqrt{5} = 5$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S74-B3-Q5

Grade 8 • Geometry • 8.G.B.8 • Set 74 • BACKWARD 3 • Q5/5

SET 74 • HSG-GPE.B.7

FORWARD 1

Find a missing vertex from an area • High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7

Triangle with vertices (0, 0), (6, 0), (2, k) has area 18. Find all possible k.

k = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S74-F1-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 74 • FORWARD 1 • Q1/5

A rectangle has vertices (0, 0), (a, 0), (a, 5), (0, 5), $a > 0$, and area 35. Find a.

a = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S74-F1-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 74 • FORWARD 1 • Q2/5

True or false?

Moving the top vertex of a triangle parallel to its base does not change its area.

A. True

B. False

Triangle (0, 0), (4, 0), (x, 6). For which x is the area 12?

A. Any x

B. No x

C. x = 4 only

D. x = 2 only

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S74-F1-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 74 • FORWARD 1 • Q4/5

Triangle (1, 1), (7, 1), (4, k) has area 21 and k > 1. Find k.

k = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S74-F1-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 74 • FORWARD 1 • Q5/5

SET 74 • HSG-GPE.B.7

FORWARD 2

Use area from coordinates with density • High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2

A park on a map has corners (0, 0), (4, 0), (4, 3), (0, 3), with 1 unit = 1 km. 1,800 people live there. Find the population density.

Density: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S74-F2-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 74 • FORWARD 2 • Q1/5

A triangular field has vertices (0, 0), (200, 0), (0, 150) in meters. Seeds are planted at 4 per m². How many seeds?

Seeds: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S74-F2-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 74 • FORWARD 2 • Q2/5

True or false?

Density based on area is an area divided by a quantity.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S74-F2-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 74 • FORWARD 2 • Q3/5

A lot with corners (0, 0), (6, 0), (6, 5), (0, 5) (meters) holds 60 plants. Density?

A. 2 plants per m²

B. 30 plants per m²

C. 11 plants per m²

D. 0.5 plants per m²

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S74-F2-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 74 • FORWARD 2 • Q4/5

A rectangle with corners (0, 0), (3, 0), (3, 2), (0, 2) (miles) has 4,200 residents. Find the density.

Density: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S74-F2-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 74 • FORWARD 2 • Q5/5

SET 75 • HSG-GMD.A.1 • GEOMETRIC MEASUREMENT AND DIMENSION

Informal arguments for the circumference and area of a circle

MAIN	Geometry main question	**HSG-GMD.A.1**
BACKWARD 1	Circumference and area formulas	**7.G.B.4**
BACKWARD 2	Area of a parallelogram	**6.G.A.1**
BACKWARD 3	All circles are similar	**HSG-C.A.1**
FORWARD 1	Area of a circle as a limit of regular polygons	**HSG-GMD.A.1**
FORWARD 2	From area to volume: stacking disks	**HSG-GMD.A.1**

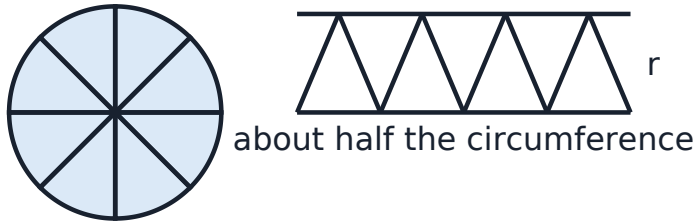
SET 75 • HSG-GMD.A.1

MAIN

Geometry main question • High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1

A circle of radius r is cut into many equal sectors, which are rearranged as shown. Explain why this shows that the area of the circle is πr^2.

Work space



Explanation: _____

Not drawn to scale.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S75-M-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 75 • MAIN • Q1/5

Explain why the ratio of circumference to diameter is the same number (π) for every circle, so that $C = \pi d = 2\pi r$.

Work space

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S75-M-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 75 • MAIN • Q2/5

A circle of radius r is cut into many equal sectors, which are rearranged, alternating point up and point down, into a near-parallelogram. Its base gets closer and closer to

A. πr^2

B. $2\pi r$ (the whole circumference)

C. r

D. πr (half the circumference)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S75-M-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 75 • MAIN • Q3/5

A regular hexagon is inscribed in a circle of radius 5. Find the hexagon's perimeter and compare it with the circle's circumference. What happens as the number of sides increases?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S75-M-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 75 • MAIN • Q4/5

True or false?

As the number of sectors increases, the rearranged shape gets closer to a rectangle with height r.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S75-M-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 75 • MAIN • Q5/5

SET 75 • HSG-GMD.A.1

BACKWARD 1

Circumference and area formulas • Grade 7 • Geometry • 7.G.B.4

Find the exact circumference of a circle of radius 4.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S75-B1-Q1

Grade 7 • Geometry • 7.G.B.4 • Set 75 • BACKWARD 1 • Q1/5

Find the exact area of a circle of radius 4.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S75-B1-Q2

Grade 7 • Geometry • 7.G.B.4 • Set 75 • BACKWARD 1 • Q2/5

Half the circumference of a circle of radius r is

A. πr^2

B. πr

C. $2\pi r$

D. $r/2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S75-B1-Q3

Grade 7 • Geometry • 7.G.B.4 • Set 75 • BACKWARD 1 • Q3/5

True or false?

$C = 2\pi r$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S75-B1-Q4

Grade 7 • Geometry • 7.G.B.4 • Set 75 • BACKWARD 1 • Q4/5

True or false?

$A = 2\pi r$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S75-B1-Q5

Grade 7 • Geometry • 7.G.B.4 • Set 75 • BACKWARD 1 • Q5/5

SET 75 • HSG-GMD.A.1

BACKWARD 2

Area of a parallelogram • Grade 6 • Geometry • 6.G.A.1

Area of a parallelogram with base 9 and height 4?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S75-B2-Q1

Grade 6 • Geometry • 6.G.A.1 • Set 75 • BACKWARD 2 • Q1/5

True or false?

A parallelogram has the same area as a rectangle with the same base and height.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S75-B2-Q2

Grade 6 • Geometry • 6.G.A.1 • Set 75 • BACKWARD 2 • Q2/5

A parallelogram's area is

A. base × slanted side

B. base × height

C. base + height

D. ½ base × height

A parallelogram has base πr and height r. Area?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S75-B2-Q4

Grade 6 • Geometry • 6.G.A.1 • Set 75 • BACKWARD 2 • Q4/5

True or false?

The slanted side is the height of a parallelogram.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S75-B2-Q5

Grade 6 • Geometry • 6.G.A.1 • Set 75 • BACKWARD 2 • Q5/5

SET 75 • HSG-GMD.A.1

BACKWARD 3

All circles are similar • High School — Geometry • Geometry: Circles • HSG-C.A.1

True or false?

Any circle can be mapped to any other circle by a translation and a dilation.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S75-B3-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.1 • Set 75 • BACKWARD 3 • Q1/5

A circle of radius 2 is dilated to radius 6. By what factor is the circumference multiplied?

Factor: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S75-B3-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.1 • Set 75 • BACKWARD 3 • Q2/5

Circles of radii 3 and 12 have circumferences in the ratio

A. 1 : 16

B. 3 : 4

C. 1 : 4

D. 1 : 9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S75-B3-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.1 • Set 75 • BACKWARD 3 • Q3/5

Why does C/d not depend on the size of the circle?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S75-B3-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.1 • Set 75 • BACKWARD 3 • Q4/5

True or false?

A larger circle has a larger value of π.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S75-B3-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.1 • Set 75 • BACKWARD 3 • Q5/5

SET 75 • HSG-GMD.A.1

FORWARD 1

Area of a circle as a limit of regular polygons • High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1

A regular n-gon is inscribed in a circle of radius r. Its area is ½ · perimeter · apothem. Explain why this approaches πr^2 as n increases.

Work space

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S75-F1-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 75 • FORWARD 1 • Q1/5

Find the area of a regular hexagon with side 2 and apothem $\sqrt{3}$. Compare with the area of its circumscribed circle (radius 2).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S75-F1-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 75 • FORWARD 1 • Q2/5

True or false?

An inscribed polygon always has less area than its circle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S75-F1-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 75 • FORWARD 1 • Q3/5

As n grows, the apothem of an inscribed regular n-gon approaches

A. πr

B. r

C. 0

D. 2r

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S75-F1-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 75 • FORWARD 1 • Q4/5

Why do mathematicians call this an "informal" argument?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S75-F1-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 75 • FORWARD 1 • Q5/5

SET 75 • HSG-GMD.A.1

FORWARD 2

From area to volume: stacking disks • High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1

A stack of 40 identical coins, each a disk of radius 1 cm and thickness 0.2 cm, forms a cylinder. Find the exact volume.

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S75-F2-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 75 • FORWARD 2 • Q1/5

True or false?

A cylinder can be thought of as a stack of congruent disks.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S75-F2-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 75 • FORWARD 2 • Q2/5

The volume of a stack of disks of area πr^2 and total height h is

A. $\frac{1}{3}\pi r^2 h$

B. $2\pi rh$

C. $\pi r^2 h$

D. πr^2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S75-F2-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 75 • FORWARD 2 • Q3/5

Explain how the area of a circle is used to find the volume of a cylinder.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S75-F2-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 75 • FORWARD 2 • Q4/5

Each disk in a stack has area 12 cm² and the stack is 5 cm high. Volume?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S75-F2-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 75 • FORWARD 2 • Q5/5

SET 76 • HSG-GMD.A.1 • GEOMETRIC MEASUREMENT AND DIMENSION

Informal argument for the volume of a cylinder

MAIN	Geometry main question	**HSG-GMD.A.1**
BACKWARD 1	Volume of a rectangular prism as layers	**5.MD.C.5**
BACKWARD 2	Area of a circle	**7.G.B.4**
BACKWARD 3	Cross-sections parallel to the base	**7.G.A.3**
FORWARD 1	Cavalieri's principle with different shapes	**HSG-GMD.A.1**
FORWARD 2	Use the cylinder volume formula to solve problems	**HSG-GMD.A.3**

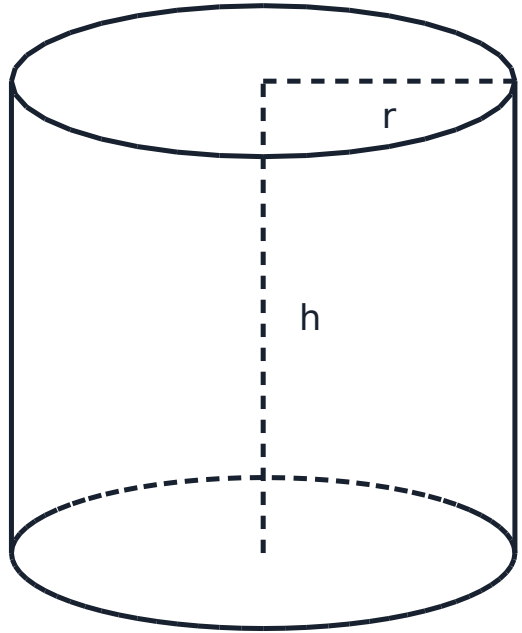
SET 76 • HSG-GMD.A.1

MAIN

Geometry main question • High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1

Explain why the volume of the cylinder shown is $V = \pi r^2 h$.

Work space



Not drawn to scale.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S76-M-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 76 • MAIN • Q1/5

A right cylinder and an oblique cylinder have the same radius r and the same height h. Explain why they have the same volume.

Work space

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S76-M-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 76 • MAIN • Q2/5

A stack of 50 coins, each of radius 1 cm and thickness 0.2 cm, has volume

A. 10π cm^3

B. 0.2π cm^3

C. 50π cm^3

D. 100π cm^3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S76-M-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 76 • MAIN • Q3/5

Explain why V = Bh works for both prisms and cylinders.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S76-M-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 76 • MAIN • Q4/5

True or false?

Pushing a straight stack of coins into a slanted stack changes its volume.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S76-M-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 76 • MAIN • Q5/5

SET 76 • HSG-GMD.A.1

BACKWARD 1

Volume of a rectangular prism as layers • Grade 5 • Measurement & Data • 5.MD.C.5

A prism has a 4 × 3 base and height 5. How many unit cubes are in one layer, and what is the volume?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S76-B1-Q1

Grade 5 • Measurement & Data • 5.MD.C.5 • Set 76 • BACKWARD 1 • Q1/5

True or false?

V = Bh, where B is the area of the base.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S76-B1-Q2

Grade 5 • Measurement & Data • 5.MD.C.5 • Set 76 • BACKWARD 1 • Q2/5

A prism has base area 20 and height 7. Volume?

A. 27

B. 140

C. 200

D. 70

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S76-B1-Q3

Grade 5 • Measurement & Data • 5.MD.C.5 • Set 76 • BACKWARD 1 • Q3/5

A prism has 6 layers of 10 cubes each. Volume?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S76-B1-Q4

Grade 5 • Measurement & Data • 5.MD.C.5 • Set 76 • BACKWARD 1 • Q4/5

True or false?

Volume is measured in square units.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S76-B1-Q5

Grade 5 • Measurement & Data • 5.MD.C.5 • Set 76 • BACKWARD 1 • Q5/5

SET 76 • HSG-GMD.A.1

BACKWARD 2

Area of a circle • Grade 7 • Geometry • 7.G.B.4

Exact area of a circle of radius 3?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S76-B2-Q1

Grade 7 • Geometry • 7.G.B.4 • Set 76 • BACKWARD 2 • Q1/5

Exact area of a circle of diameter 10?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S76-B2-Q2

Grade 7 • Geometry • 7.G.B.4 • Set 76 • BACKWARD 2 • Q2/5

A =

A. $2\pi r$

B. πr

C. πr^2

D. πd

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S76-B2-Q3

Grade 7 • Geometry • 7.G.B.4 • Set 76 • BACKWARD 2 • Q3/5

True or false?

The base of a cylinder is a circle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S76-B2-Q4

Grade 7 • Geometry • 7.G.B.4 • Set 76 • BACKWARD 2 • Q4/5

True or false?

The area of a circle of radius 2 is 4.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S76-B2-Q5

Grade 7 • Geometry • 7.G.B.4 • Set 76 • BACKWARD 2 • Q5/5

SET 76 • HSG-GMD.A.1

BACKWARD 3

Cross-sections parallel to the base • Grade 7 • Geometry • 7.G.A.3

True or false?

A slice of a right prism parallel to its base is congruent to the base.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S76-B3-Q1

Grade 7 • Geometry • 7.G.A.3 • Set 76 • BACKWARD 3 • Q1/5

A cylinder is sliced parallel to its base. Describe the cross-section.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S76-B3-Q2

Grade 7 • Geometry • 7.G.A.3 • Set 76 • BACKWARD 3 • Q2/5

A rectangular prism sliced parallel to its 3 × 4 base gives a

A. smaller rectangle

B. 3 × 4 rectangle

C. triangle

D. circle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S76-B3-Q3

Grade 7 • Geometry • 7.G.A.3 • Set 76 • BACKWARD 3 • Q3/5

Why do all horizontal slices of a cylinder have the same area?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S76-B3-Q4

Grade 7 • Geometry • 7.G.A.3 • Set 76 • BACKWARD 3 • Q4/5

True or false?

Horizontal slices of a cylinder get smaller toward the top.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S76-B3-Q5

Grade 7 • Geometry • 7.G.A.3 • Set 76 • BACKWARD 3 • Q5/5

SET 76 • HSG-GMD.A.1

FORWARD 1

Cavalieri's principle with different shapes • High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1

A cylinder with base area 50 cm² and a square prism with base area 50 cm² both have height 8 cm. Compare their volumes.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S76-F1-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 76 • FORWARD 1 • Q1/5

An oblique cylinder has radius 3 and vertical height 7 (its slanted side is 9). Find its exact volume.

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S76-F1-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 76 • FORWARD 1 • Q2/5

True or false?

Cavalieri's principle requires equal cross-sectional areas at every height and equal heights.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S76-F1-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 76 • FORWARD 1 • Q3/5

Two stacks of 30 identical cards, one straight and one slanted, have

A. different volumes

B. equal volumes

C. equal surface areas

D. no volume

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S76-F1-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 76 • FORWARD 1 • Q4/5

State Cavalieri's principle in your own words.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S76-F1-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 76 • FORWARD 1 • Q5/5

SET 76 • HSG-GMD.A.1

FORWARD 2

Use the cylinder volume formula to solve problems • High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3

A can has radius 4 cm and height 12 cm. Find its volume to the nearest cubic centimeter.

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S76-F2-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 76 • FORWARD 2 • Q1/5

A pipe 2 m long has inner radius 0.1 m. Find the volume of water it holds, to the nearest thousandth of a cubic meter.

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S76-F2-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 76 • FORWARD 2 • Q2/5

True or false?

Doubling the height of a cylinder multiplies its volume by 4.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S76-F2-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 76 • FORWARD 2 • Q3/5

Doubling the radius of a cylinder multiplies its volume by

A. 4

B. 8

C. 6

D. 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S76-F2-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 76 • FORWARD 2 • Q4/5

A cylindrical tank has diameter 6 ft and height 10 ft. Exact volume?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S76-F2-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 76 • FORWARD 2 • Q5/5

SET 77 • HSG-GMD.A.1 • GEOMETRIC MEASUREMENT AND DIMENSION

Informal arguments for the volumes of a pyramid and a cone

MAIN	Geometry main question	**HSG-GMD.A.1**
BACKWARD 1	Volume of a prism	**7.G.B.6**
BACKWARD 2	Base area of a pyramid	**6.G.A.1**
BACKWARD 3	The cylinder volume argument	**HSG-GMD.A.1**
FORWARD 1	Cavalieri's principle for pyramids and cones	**HSG-GMD.A.1**
FORWARD 2	Use pyramid and cone volume formulas	**HSG-GMD.A.3**

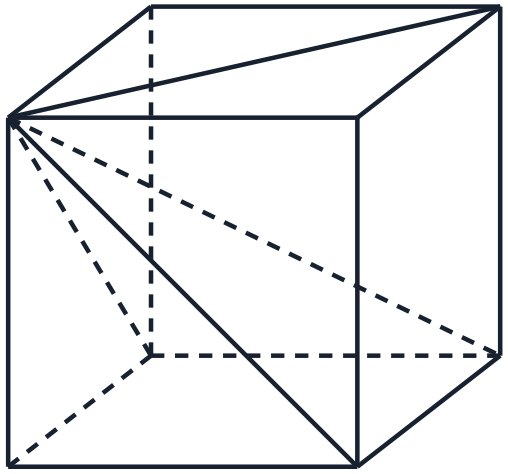
SET 77 • HSG-GMD.A.1

MAIN

Geometry main question • High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1

A cube is split into three congruent pyramids that share one vertex of the cube as their apex, as shown. Use this to explain why a pyramid's volume is ⅓Bh.

Work space



Explanation: _____

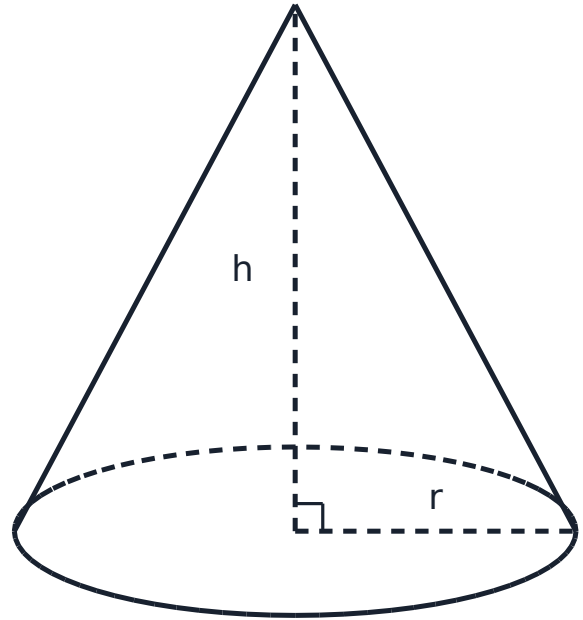
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S77-M-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 77 • MAIN • Q1/5

Explain why the volume of a cone is $\frac{1}{3}\pi r^2 h$, using pyramids.

Work space



Not drawn to scale.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S77-M-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 77 • MAIN • Q2/5

A cone and a cylinder have the same base and the same height. How many cones full of water fill the cylinder?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S77-M-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 77 • MAIN • Q3/5

A cube with side 6 is split into three congruent pyramids that share one vertex of the cube as their apex. The volume of each pyramid is

A. 36

B. 216

C. 108

D. 72

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S77-M-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 77 • MAIN • Q4/5

True or false?

A pyramid has half the volume of a prism with the same base and height.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S77-M-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 77 • MAIN • Q5/5

SET 77 • HSG-GMD.A.1

BACKWARD 1

Volume of a prism • Grade 7 • Geometry • 7.G.B.6

A prism has base area 36 and height 6. Volume?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S77-B1-Q1

Grade 7 • Geometry • 7.G.B.6 • Set 77 • BACKWARD 1 • Q1/5

A cube has side 6. Volume?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S77-B1-Q2

Grade 7 • Geometry • 7.G.B.6 • Set 77 • BACKWARD 1 • Q2/5

V =

A. Bh

B. ⅓Bh

C. 2Bh

D. B + h

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S77-B1-Q3

Grade 7 • Geometry • 7.G.B.6 • Set 77 • BACKWARD 1 • Q3/5

True or false?

A cube is a prism.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S77-B1-Q4

Grade 7 • Geometry • 7.G.B.6 • Set 77 • BACKWARD 1 • Q4/5

True or false?

A prism's volume depends on the number of its faces.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S77-B1-Q5

Grade 7 • Geometry • 7.G.B.6 • Set 77 • BACKWARD 1 • Q5/5

SET 77 • HSG-GMD.A.1

BACKWARD 2

Base area of a pyramid • Grade 6 • Geometry • 6.G.A.1

A square base has side 10. Area?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S77-B2-Q1

Grade 6 • Geometry • 6.G.A.1 • Set 77 • BACKWARD 2 • Q1/5

A triangular base has base 6 and height 4. Area?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S77-B2-Q2

Grade 6 • Geometry • 6.G.A.1 • Set 77 • BACKWARD 2 • Q2/5

The base of a cone is a

A. triangle

B. circle

C. sector

D. square

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S77-B2-Q3

Grade 6 • Geometry • 6.G.A.1 • Set 77 • BACKWARD 2 • Q3/5

True or false?

B in $V = \frac{1}{3}Bh$ means the area of the base.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S77-B2-Q4

Grade 6 • Geometry • 6.G.A.1 • Set 77 • BACKWARD 2 • Q4/5

True or false?

B is the length of a base edge.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S77-B2-Q5

Grade 6 • Geometry • 6.G.A.1 • Set 77 • BACKWARD 2 • Q5/5

SET 77 • HSG-GMD.A.1

BACKWARD 3

The cylinder volume argument • High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1

True or false?

A cylinder's volume is $\pi r^2 h$ because it is a stack of disks of area πr^2.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S77-B3-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 77 • BACKWARD 3 • Q1/5

Cylinder radius 3, height 5: exact volume?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S77-B3-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 77 • BACKWARD 3 • Q2/5

A cone fits inside a cylinder with the same base and height. The cone's volume is

A. equal to the cylinder's

B. ⅔ of the cylinder's

C. ½ of the cylinder's

D. ⅓ of the cylinder's

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S77-B3-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 77 • BACKWARD 3 • Q3/5

A cylinder holds 60 cm³. A cone with the same base and height holds

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S77-B3-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 77 • BACKWARD 3 • Q4/5

True or false?

A cone is a cylinder with the top cut off flat.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S77-B3-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 77 • BACKWARD 3 • Q5/5

SET 77 • HSG-GMD.A.1

FORWARD 1

Cavalieri's principle for pyramids and cones • High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1

A square pyramid and a cone both have base area 50 cm² and height 9 cm. Compare their volumes.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S77-F1-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 77 • FORWARD 1 • Q1/5

An oblique cone has radius 4 and vertical height 6. Find its exact volume.

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S77-F1-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 77 • FORWARD 1 • Q2/5

True or false?

Two pyramids with equal base areas and equal heights have equal volumes even if their apexes are in different positions.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S77-F1-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 77 • FORWARD 1 • Q3/5

Why do a pyramid and a cone with equal base areas and heights have equal cross-section areas at every level?

A. Their bases have the same shape

B. Both cross-sections are scaled copies of the base by the same factor

C. They have the same slant height

D. They have the same apex

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S77-F1-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 77 • FORWARD 1 • Q4/5

A cross-section of a pyramid halfway up has what fraction of the base area?

Fraction: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S77-F1-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 77 • FORWARD 1 • Q5/5

SET 77 • HSG-GMD.A.1

FORWARD 2

Use pyramid and cone volume formulas • High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3

A square pyramid has base side 9 m and height 10 m. Volume?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S77-F2-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 77 • FORWARD 2 • Q1/5

A cone has radius 3 in. and height 8 in. Find its volume to the nearest tenth.

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S77-F2-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 77 • FORWARD 2 • Q2/5

True or false?

A cone with the same base and height as a cylinder of volume 90 has volume 45.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S77-F2-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 77 • FORWARD 2 • Q3/5

A pyramid with base area 48 and height 5 has volume

A. 120

B. 53

C. 240

D. 80

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S77-F2-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 77 • FORWARD 2 • Q4/5

A cone-shaped cup has diameter 6 cm and height 9 cm. Exact volume?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S77-F2-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 77 • FORWARD 2 • Q5/5

SET 78 • HSG-GMD.A.3 • GEOMETRIC MEASUREMENT AND DIMENSION

Use volume formulas for cylinders, pyramids, and cones

MAIN	Geometry main question	**HSG-GMD.A.3**
BACKWARD 1	Area of a circle	**7.G.B.4**
BACKWARD 2	Find a missing side with the Pythagorean Theorem	**8.G.B.7**
BACKWARD 3	Evaluate a formula and round	**7.EE.B.3**
BACKWARD 4	Which solid uses ⅓?	**HSG-GMD.A.1**
FORWARD 1	Composite and hollow solids	**HSG-GMD.A.3**
FORWARD 2	Mass from volume and density	**HSG-MG.A.2**

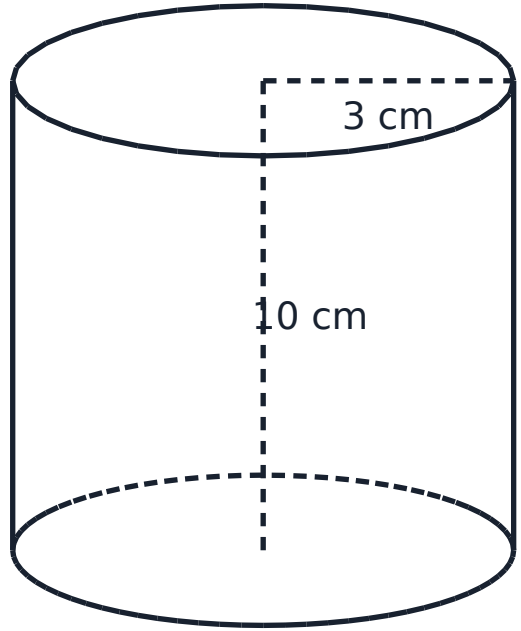
SET 78 • HSG-GMD.A.3

MAIN

Geometry main question • High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3

A cylinder has radius 3 cm and height 10 cm. Find its volume, exact and to the nearest tenth.

Volume: _____



Not drawn to scale.

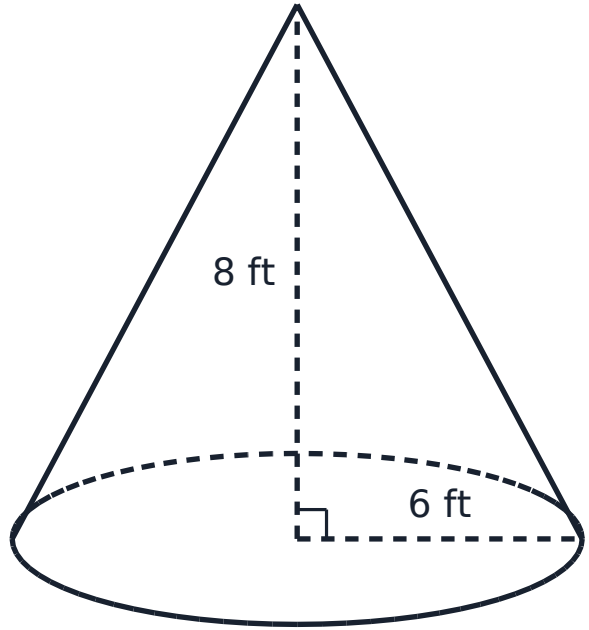
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S78-M-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 78 • MAIN • Q1/5

A cone has radius 6 ft and height 8 ft. Find its volume, exact and to the nearest tenth.

Volume: _____



Not drawn to scale.

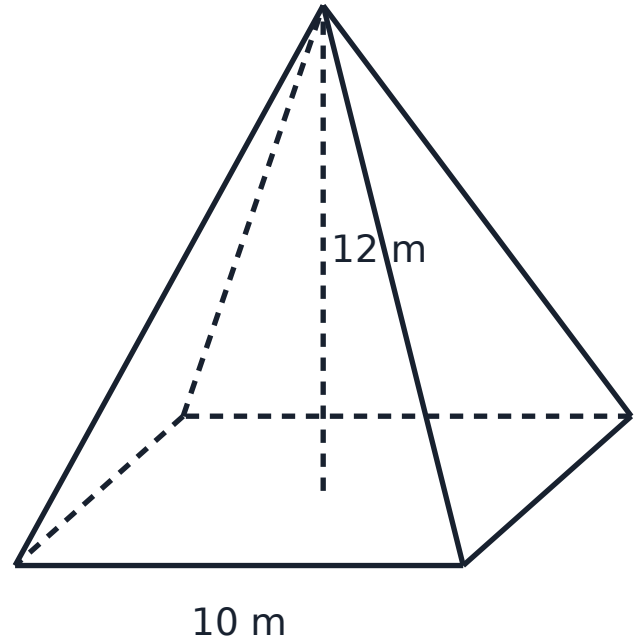
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S78-M-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 78 • MAIN • Q2/5

A square pyramid has base side 10 m and height 12 m. Find its volume.

Volume: _____



Not drawn to scale.

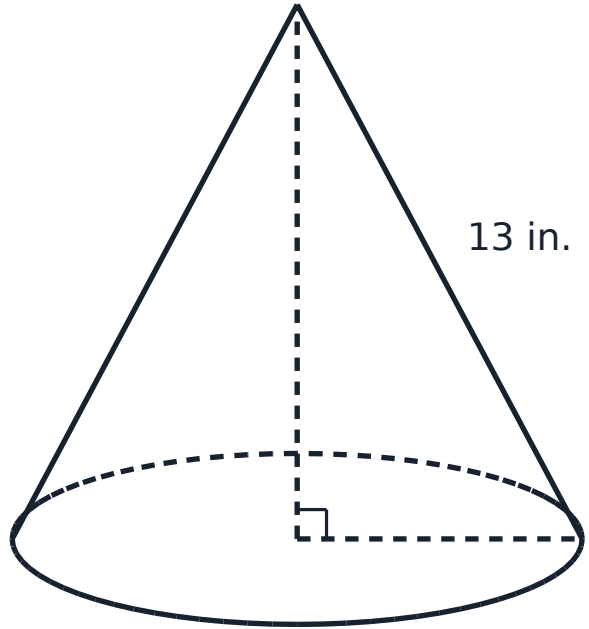
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S78-M-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 78 • MAIN • Q3/5

A cone has diameter 10 in. and slant height 13 in. Find its volume, exact and to the nearest tenth.

Volume: _____



Not drawn to scale.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S78-M-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 78 • MAIN • Q4/5

A cylinder and a cone have the same radius and height. The cylinder holds 600 mL. The cone holds

A. 300 mL

B. 200 mL

C. 600 mL

D. 1,800 mL

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S78-M-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 78 • MAIN • Q5/5

SET 78 • HSG-GMD.A.3

BACKWARD 1

Area of a circle • Grade 7 • Geometry • 7.G.B.4

Exact area of a circle with radius 6?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S78-B1-Q1

Grade 7 • Geometry • 7.G.B.4 • Set 78 • BACKWARD 1 • Q1/5

Exact area of a circle with diameter 10?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S78-B1-Q2

Grade 7 • Geometry • 7.G.B.4 • Set 78 • BACKWARD 1 • Q2/5

A base of radius 3 has area

A. 9

B. 9π

C. 3π

D. 6π

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S78-B1-Q3

Grade 7 • Geometry • 7.G.B.4 • Set 78 • BACKWARD 1 • Q3/5

True or false?

Squaring the radius comes before multiplying by π.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S78-B1-Q4

Grade 7 • Geometry • 7.G.B.4 • Set 78 • BACKWARD 1 • Q4/5

True or false?

πr^2 with $r = 3$ is $(3\pi)^2$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S78-B1-Q5

Grade 7 • Geometry • 7.G.B.4 • Set 78 • BACKWARD 1 • Q5/5

SET 78 • HSG-GMD.A.3

BACKWARD 2

Find a missing side with the Pythagorean Theorem • Grade 8 • Geometry • 8.G.B.7

A right triangle has hypotenuse 13 and one leg 5. Other leg?

Leg: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S78-B2-Q1

Grade 8 • Geometry • 8.G.B.7 • Set 78 • BACKWARD 2 • Q1/5

A cone has radius 6 and slant height 10. Height?

Height: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S78-B2-Q2

Grade 8 • Geometry • 8.G.B.7 • Set 78 • BACKWARD 2 • Q2/5

Radius, height, and slant height of a cone form a

A. square

B. right triangle

C. equilateral triangle

D. circle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S78-B2-Q3

Grade 8 • Geometry • 8.G.B.7 • Set 78 • BACKWARD 2 • Q3/5

True or false?

The slant height is longer than the height.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S78-B2-Q4

Grade 8 • Geometry • 8.G.B.7 • Set 78 • BACKWARD 2 • Q4/5

True or false?

The slant height can be used as the height in $V = \frac{1}{3}\pi r^2 h$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S78-B2-Q5

Grade 8 • Geometry • 8.G.B.7 • Set 78 • BACKWARD 2 • Q5/5

SET 78 • HSG-GMD.A.3

BACKWARD 3

Evaluate a formula and round • Grade 7 • Expressions & Equations • 7.EE.B.3

Evaluate ⅓ · 36 · 8.

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S78-B3-Q1

Grade 7 • Expressions & Equations • 7.EE.B.3 • Set 78 • BACKWARD 3 • Q1/5

Round 90π to the nearest tenth.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S78-B3-Q2

Grade 7 • Expressions & Equations • 7.EE.B.3 • Set 78 • BACKWARD 3 • Q2/5

$96\pi \approx$

A. 301.6

B. 301.5

C. 3,016

D. 96.3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S78-B3-Q3

Grade 7 • Expressions & Equations • 7.EE.B.3 • Set 78 • BACKWARD 3 • Q3/5

True or false?

Using 3.14 for π may change the last digit of a rounded answer.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S78-B3-Q4

Grade 7 • Expressions & Equations • 7.EE.B.3 • Set 78 • BACKWARD 3 • Q4/5

True or false?

$\frac{1}{3} \cdot 100 \cdot 12 = 1{,}200$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S78-B3-Q5

Grade 7 • Expressions & Equations • 7.EE.B.3 • Set 78 • BACKWARD 3 • Q5/5

SET 78 • HSG-GMD.A.3

BACKWARD 4

Which solid uses ⅓? • High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1

True or false?

Pyramids and cones use $V = \frac{1}{3}Bh$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S78-B4-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 78 • BACKWARD 4 • Q1/5

Which has the formula V = Bh: a cone or a cylinder?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S78-B4-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 78 • BACKWARD 4 • Q2/5

$V = \frac{1}{3}\pi r^2 h$ is the volume of a

A. cylinder

B. cone

C. prism

D. sphere

Why does a cone have ⅓ of the matching cylinder's volume?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S78-B4-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 78 • BACKWARD 4 • Q4/5

True or false?

A pyramid uses $V = Bh$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S78-B4-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 78 • BACKWARD 4 • Q5/5

SET 78 • HSG-GMD.A.3

FORWARD 1

Composite and hollow solids • High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3

A silo is a cylinder (radius 4 m, height 10 m) topped by a cone (radius 4 m, height 3 m). Find its exact volume.

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S78-F1-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 78 • FORWARD 1 • Q1/5

A pipe is a cylinder of length 20 cm with outer radius 5 cm and inner radius 4 cm. Find the volume of the material, exact.

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S78-F1-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 78 • FORWARD 1 • Q2/5

True or false?

The volume of a composite solid is the sum of the volumes of its non-overlapping parts.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S78-F1-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 78 • FORWARD 1 • Q3/5

A cylinder (V = 300) has a cone with the same base and height carved out. Remaining volume?

A. 300

B. 200

C. 0

D. 100

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S78-F1-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 78 • FORWARD 1 • Q4/5

A pencil tip is a cone of radius 0.4 cm and height 1.5 cm on a cylinder of radius 0.4 cm and length 15 cm. Find the total volume to the nearest tenth.

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S78-F1-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 78 • FORWARD 1 • Q5/5

SET 78 • HSG-GMD.A.3

FORWARD 2

Mass from volume and density • High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2

A steel cylinder has radius 2 cm and height 5 cm. Steel has density 7.85 g/cm³. Find the mass to the nearest gram.

Mass: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S78-F2-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 78 • FORWARD 2 • Q1/5

A cone of ice cream (radius 3 cm, height 10 cm) has density 0.55 g/cm³. Find the mass to the nearest gram.

Mass: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S78-F2-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 78 • FORWARD 2 • Q2/5

True or false?

Mass = volume ÷ density.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S78-F2-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 78 • FORWARD 2 • Q3/5

A pyramid of volume 400 cm³ is made of material with density 2 g/cm³. Mass?

A. 200 g

B. 402 g

C. 1,600 g

D. 800 g

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S78-F2-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 78 • FORWARD 2 • Q4/5

A gold cube has volume 8 cm³ and mass 154.4 g. Find the density.

Density: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S78-F2-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 78 • FORWARD 2 • Q5/5

SET 79 • HSG-GMD.A.3 • GEOMETRIC MEASUREMENT AND DIMENSION

Use the volume formula for spheres, including composite solids

MAIN	Geometry main question	**HSG-GMD.A.3**
BACKWARD 1	Cubes of numbers	**6.EE.A.1**
BACKWARD 2	Radius from diameter	**7.G.B.4**
BACKWARD 3	Multiply fractions by numbers	**7.NS.A.2**
BACKWARD 4	Cylinder and cone volumes	**HSG-GMD.A.3**
FORWARD 1	Scaling spheres	**HSG-GMD.A.3**
FORWARD 2	Model real objects as spheres	**HSG-MG.A.1**

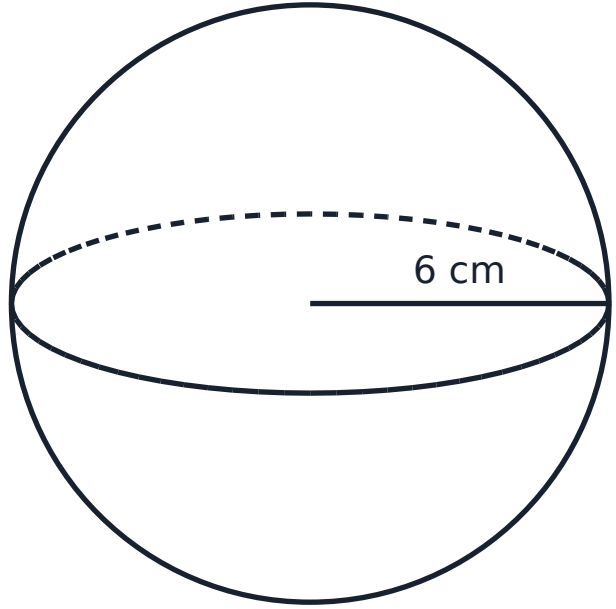
SET 79 • HSG-GMD.A.3

MAIN

Geometry main question • High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3

Find the volume of a sphere with radius 6 cm, exact and to the nearest tenth.

Volume: _____



Not drawn to scale.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S79-M-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 79 • MAIN • Q1/5

Find the volume of a ball with diameter 10 in., exact and to the nearest tenth.

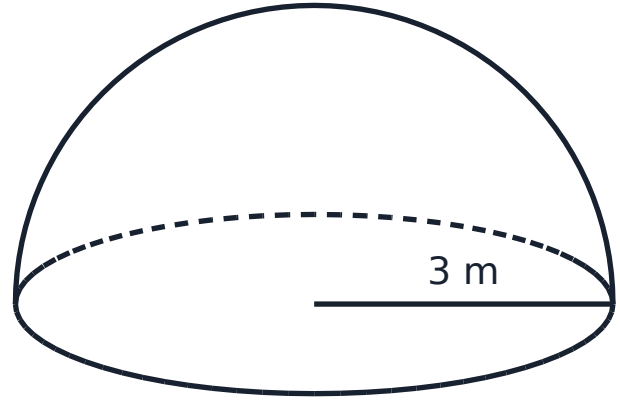
Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S79-M-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 79 • MAIN • Q2/5

Find the volume of a hemisphere with radius 3 m, exact and to the nearest tenth.



Not drawn to scale.

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S79-M-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 79 • MAIN • Q3/5

A capsule is a cylinder of radius 3 cm and length 10 cm with a hemisphere on each end. Find its volume, exact and to the nearest tenth.

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S79-M-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 79 • MAIN • Q4/5

An ice cream cone (radius 2, height 9) is topped by a hemisphere of radius 2. The total exact volume is

A. 52π/3

B. 28π

C. 12π

D. 44π/3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S79-M-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 79 • MAIN • Q5/5

SET 79 • HSG-GMD.A.3

BACKWARD 1

Cubes of numbers • Grade 6 • Expressions & Equations • 6.EE.A.1

$6^3 =$

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S79-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 79 • BACKWARD 1 • Q1/5

$5^3 =$

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S79-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 79 • BACKWARD 1 • Q2/5

$3^3 =$

A. 9

B. 81

C. 6

D. 27

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S79-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 79 • BACKWARD 1 • Q3/5

True or false?

$2^3 = 8$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S79-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 79 • BACKWARD 1 • Q4/5

True or false?

$4^3 = 12$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S79-B1-Q5

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 79 • BACKWARD 1 • Q5/5

SET 79 • HSG-GMD.A.3

BACKWARD 2

Radius from diameter • Grade 7 • Geometry • 7.G.B.4

A ball has diameter 10. Radius?

Radius: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S79-B2-Q1

Grade 7 • Geometry • 7.G.B.4 • Set 79 • BACKWARD 2 • Q1/5

True or false?

Using the diameter in 4/3 πr³ makes the answer 8 times too large.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S79-B2-Q2

Grade 7 • Geometry • 7.G.B.4 • Set 79 • BACKWARD 2 • Q2/5

A sphere with diameter 12 has r^3 =

A. 216

B. 144

C. 36

D. 1,728

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S79-B2-Q3

Grade 7 • Geometry • 7.G.B.4 • Set 79 • BACKWARD 2 • Q3/5

A sphere has radius 2.5. Diameter?

Diameter: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S79-B2-Q4

Grade 7 • Geometry • 7.G.B.4 • Set 79 • BACKWARD 2 • Q4/5

True or false?

The radius of a sphere is twice its diameter.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S79-B2-Q5

Grade 7 • Geometry • 7.G.B.4 • Set 79 • BACKWARD 2 • Q5/5

SET 79 • HSG-GMD.A.3

BACKWARD 3

Multiply fractions by numbers • Grade 7 • The Number System • 7.NS.A.2

$4/3 \cdot 216 =$

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S79-B3-Q1

Grade 7 • The Number System • 7.NS.A.2 • Set 79 • BACKWARD 3 • Q1/5

½ · 4/3 · 27 =

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S79-B3-Q2

Grade 7 • The Number System • 7.NS.A.2 • Set 79 • BACKWARD 3 • Q2/5

$4/3 \cdot 125 =$

A. 375/4

B. 500/3

C. 125/3

D. 500

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S79-B3-Q3

Grade 7 • The Number System • 7.NS.A.2 • Set 79 • BACKWARD 3 • Q3/5

True or false?

$4/3 \cdot 27 = 36$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S79-B3-Q4

Grade 7 • The Number System • 7.NS.A.2 • Set 79 • BACKWARD 3 • Q4/5

True or false?

$4/3 \cdot 6 = 24$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S79-B3-Q5

Grade 7 • The Number System • 7.NS.A.2 • Set 79 • BACKWARD 3 • Q5/5

SET 79 • HSG-GMD.A.3

BACKWARD 4

Cylinder and cone volumes • High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3

Cylinder radius 3, length 10: exact volume?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S79-B4-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 79 • BACKWARD 4 • Q1/5

Cone radius 2, height 9: exact volume?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S79-B4-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 79 • BACKWARD 4 • Q2/5

A cylinder and a cone with radius 3 and height 4 have volumes

A. 36π and 12π

B. 12π and 36π

C. 12π and 4π

D. 36π and 36π

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S79-B4-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 79 • BACKWARD 4 • Q3/5

True or false?

Two hemispheres of radius r make one sphere of radius r.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S79-B4-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 79 • BACKWARD 4 • Q4/5

True or false?

A composite volume is the product of the parts.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S79-B4-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 79 • BACKWARD 4 • Q5/5

SET 79 • HSG-GMD.A.3

FORWARD 1

Scaling spheres • High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3

The radius of a sphere is doubled. By what factor is the volume multiplied?

Factor: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S79-F1-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 79 • FORWARD 1 • Q1/5

A sphere fits exactly inside a cube with side 6. Find the exact volume of the space in the cube outside the sphere.

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S79-F1-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 79 • FORWARD 1 • Q2/5

True or false?

A sphere of radius 3 has 9 times the volume of a sphere of radius 1.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S79-F1-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 79 • FORWARD 1 • Q3/5

A balloon's radius grows from 10 cm to 15 cm. Its volume is multiplied by

A. 1.5

B. 2.25

C. 5

D. 3.375

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S79-F1-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 79 • FORWARD 1 • Q4/5

A sphere and a cylinder have the same radius r, and the cylinder's height is 2r. What fraction of the cylinder's volume is the sphere's?

Fraction: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S79-F1-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 79 • FORWARD 1 • Q5/5

SET 79 • HSG-GMD.A.3

FORWARD 2

Model real objects as spheres • High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1

Model Earth as a sphere with radius 6,371 km. Estimate its volume in cubic kilometers, in scientific notation to 3 significant figures.

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S79-F2-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 79 • FORWARD 2 • Q1/5

A basketball has circumference 75 cm. Model it as a sphere and find its volume to the nearest 100 cm^3.

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S79-F2-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 79 • FORWARD 2 • Q2/5

True or false?

Modeling an orange as a sphere ignores its small bumps.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S79-F2-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 79 • FORWARD 2 • Q3/5

A scoop of ice cream is modeled as a sphere of radius 3 cm. Its volume is about

A. 28 cm^3

B. 113 cm^3

C. 339 cm^3

D. 36 cm^3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S79-F2-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 79 • FORWARD 2 • Q4/5

Give one reason a sphere is a good model for a ball bearing and one limitation of the model for a pumpkin.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S79-F2-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 79 • FORWARD 2 • Q5/5

Solve for an unknown dimension from a volume

MAIN	Geometry main question	**HSG-GMD.A.3**
BACKWARD 1	Solve a one-step equation with a coefficient	**HSA-REI.B.3**
BACKWARD 2	Square roots and cube roots	**8.EE.A.2**
BACKWARD 3	Rearrange a formula	**HSA-CED.A.4**
FORWARD 1	Find dimensions to meet a target volume	**HSG-GMD.A.3**
FORWARD 2	Design a container that meets constraints	**HSG-MG.A.3**

SET 80 • HSG-GMD.A.3

MAIN

Geometry main question • High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3

A cylinder has volume 360π cm³ and radius 6 cm. Find its height.

Height: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S80-M-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 80 • MAIN • Q1/5

A cone has volume 96π in^3 and height 8 in. Find its radius.

Radius: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S80-M-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 80 • MAIN • Q2/5

A sphere has volume 36π m³. Find its radius.

Radius: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S80-M-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 80 • MAIN • Q3/5

A square pyramid has volume 256 ft³ and height 12 ft. Find the side of its base.

Side: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S80-M-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 80 • MAIN • Q4/5

A cylinder holds 500 cm³ and has height 10 cm. Its radius is closest to

A. 7.1 cm

B. 4.0 cm

C. 2.0 cm

D. 15.9 cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S80-M-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 80 • MAIN • Q5/5

SET 80 • HSG-GMD.A.3

BACKWARD 1

Solve a one-step equation with a coefficient • High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3

Solve $36\pi h = 360\pi$.

$h =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S80-B1-Q1

High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 80 • BACKWARD 1 • Q1/5

Solve ⅓x = 12.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S80-B1-Q2

High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 80 • BACKWARD 1 • Q2/5

Solve $4x = 108$.

A. 25

B. 104

C. 432

D. 27

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S80-B1-Q3**

High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 80 • BACKWARD 1 • Q3/5

True or false?

To solve $36\pi h = 360\pi$, dividing both sides by π is allowed.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S80-B1-Q4

High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 80 • BACKWARD 1 • Q4/5

True or false?

To solve ⅓x = 12, divide by 3.

A. True

B. False

SET 80 • HSG-GMD.A.3

BACKWARD 2

Square roots and cube roots • Grade 8 • Expressions & Equations • 8.EE.A.2

Solve $r^2 = 36$ for $r > 0$.

$r =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S80-B2-Q1

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 80 • BACKWARD 2 • Q1/5

Solve $r^3 = 27$.

r = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S80-B2-Q2

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 80 • BACKWARD 2 • Q2/5

$\sqrt[3]{125} =$

A. 25

B. 5

C. 41.7

D. 15

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S80-B2-Q3

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 80 • BACKWARD 2 • Q3/5

True or false?

$\sqrt{64} = 8$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S80-B2-Q4

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 80 • BACKWARD 2 • Q4/5

True or false?

If $r^3 = 8$, then $r = 4$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S80-B2-Q5

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 80 • BACKWARD 2 • Q5/5

SET 80 • HSG-GMD.A.3

BACKWARD 3

Rearrange a formula • High School — Algebra I (course placement in this bank) • Algebra: Creating Equations • HSA-CED.A.4

Solve $V = \pi r^2 h$ for h.

$h =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S80-B3-Q1

High School — Algebra I (course placement in this bank) • Algebra: Creating Equations • HSA-CED.A.4 • Set 80 • BACKWARD 3 • Q1/5

Solve $V = \frac{1}{3}Bh$ for B.

B = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S80-B3-Q2

High School — Algebra I (course placement in this bank) • Algebra: Creating Equations • HSA-CED.A.4 • Set 80 • BACKWARD 3 • Q2/5

$V = 4/3\ \pi r^3$ solved for r^3 is

A. $4V/(3\pi)$

B. $V/(4\pi)$

C. $3V/(4\pi)$

D. $3\pi V/4$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S80-B3-Q3**

High School — Algebra I (course placement in this bank) • Algebra: Creating Equations • HSA-CED.A.4 • Set 80 • BACKWARD 3 • Q3/5

True or false?

Solving $V = \pi r^2 h$ for r needs a square root.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S80-B3-Q4

High School — Algebra I (course placement in this bank) • Algebra: Creating Equations • HSA-CED.A.4 • Set 80 • BACKWARD 3 • Q4/5

True or false?

Solving $V = Bh$ for h gives $h = VB$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S80-B3-Q5

High School — Algebra I (course placement in this bank) • Algebra: Creating Equations • HSA-CED.A.4 • Set 80 • BACKWARD 3 • Q5/5

SET 80 • HSG-GMD.A.3

FORWARD 1

Find dimensions to meet a target volume • High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3

A drink can must hold 355 cm³ and have radius 3.3 cm. Find the height to the nearest tenth of a centimeter.

Height: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S80-F1-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 80 • FORWARD 1 • Q1/5

A spherical tank must hold 1,000 m³. Find its radius to the nearest tenth of a meter.

Radius: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S80-F1-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 80 • FORWARD 1 • Q2/5

True or false?

To double the volume of a cylinder while keeping its radius, double its height.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S80-F1-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 80 • FORWARD 1 • Q3/5

A cone has the same volume as a cylinder of equal radius. The cone's height is

A. 9 times the cylinder's

B. ⅓ of the cylinder's

C. 3 times the cylinder's

D. equal to the cylinder's

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S80-F1-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 80 • FORWARD 1 • Q4/5

A cube has the same volume as a sphere of radius 3. Find the cube's edge to the nearest tenth.

Edge: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S80-F1-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 80 • FORWARD 1 • Q5/5

SET 80 • HSG-GMD.A.3

FORWARD 2

Design a container that meets constraints • High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3

A cylindrical vase must hold 2,000 cm^3 and be 20 cm tall. Its radius must be at least 5 cm to fit a bouquet. Does the design work? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S80-F2-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 80 • FORWARD 2 • Q1/5

A conical paper cup must hold 150 cm³ and have radius 3.5 cm. Find the height to the nearest tenth.

Height: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S80-F2-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 80 • FORWARD 2 • Q2/5

True or false?

A container's volume requirement fixes each of its dimensions.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S80-F2-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 80 • FORWARD 2 • Q3/5

A cylinder must hold 100π cm^3. Which (r, h) works?

A. (5, 4)

B. (5, 20)

C. (4, 5)

D. (10, 10)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S80-F2-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 80 • FORWARD 2 • Q4/5

Give two different cylinders (radius and height) that each hold 64π cm^3.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S80-F2-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 80 • FORWARD 2 • Q5/5

SET 81 • HSG-GMD.B.4 • GEOMETRIC MEASUREMENT AND DIMENSION

Identify cross-sections of three-dimensional objects

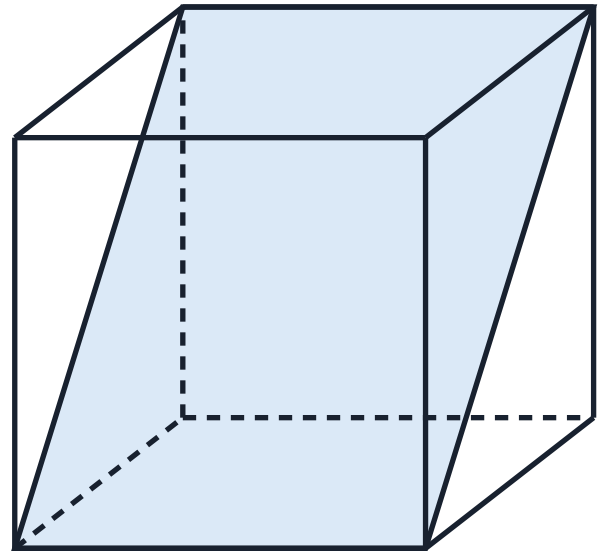
MAIN	Geometry main question	**HSG-GMD.B.4**
BACKWARD 1	Cross-sections of right rectangular prisms and pyramids	**7.G.A.3**
BACKWARD 2	Properties of polygons	**5.G.B.4**
BACKWARD 3	Parts of solids	**6.G.A.4**
FORWARD 1	Areas of cross-sections	**HSG-GMD.B.4**
FORWARD 2	Use cross-section areas to compare volumes (Cavalieri)	**HSG-GMD.A.2**

SET 81 • HSG-GMD.B.4

MAIN

Geometry main question • High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4

A plane cuts the cube as shown. Name the shape of the cross-section.



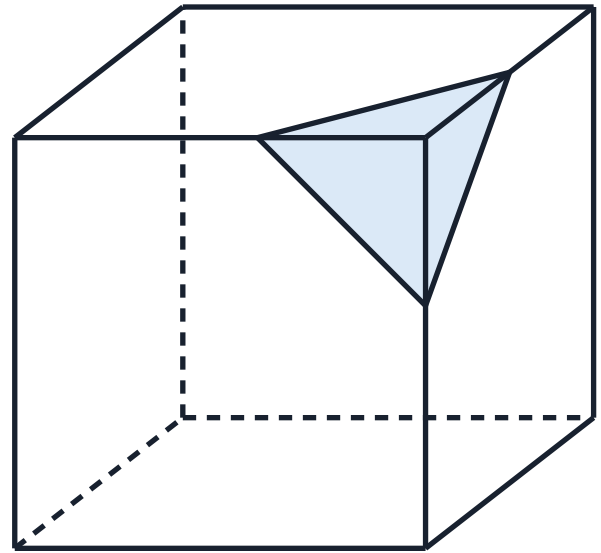
Shape: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S81-M-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 81 • MAIN • Q1/5

A plane cuts off one corner of the cube as shown. Name the shape of the cross-section.



Shape: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S81-M-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 81 • MAIN • Q2/5

A right circular cone is cut by a vertical plane through its apex. The cross-section is

A. a circle

B. a rectangle

C. a triangle (isosceles)

D. an ellipse

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S81-M-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 81 • MAIN • Q3/5

Name the cross-section of a cylinder cut by a plane (a) parallel to its base, (b) perpendicular to its base through its axis, and (c) slanted, cutting the curved side only.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S81-M-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 81 • MAIN • Q4/5

True or false?

Every plane section of a sphere that cuts through it is a circle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S81-M-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 81 • MAIN • Q5/5

SET 81 • HSG-GMD.B.4

BACKWARD 1

Cross-sections of right rectangular prisms and pyramids • Grade 7 • Geometry • 7.G.A.3

A rectangular prism is cut parallel to its base. Shape?

Shape: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S81-B1-Q1

Grade 7 • Geometry • 7.G.A.3 • Set 81 • BACKWARD 1 • Q1/5

A square pyramid is cut parallel to its base. Shape?

Shape: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S81-B1-Q2

Grade 7 • Geometry • 7.G.A.3 • Set 81 • BACKWARD 1 • Q2/5

A square pyramid cut vertically through its apex and two opposite base edges' midpoints gives a

A. trapezoid

B. square

C. circle

D. triangle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S81-B1-Q3

Grade 7 • Geometry • 7.G.A.3 • Set 81 • BACKWARD 1 • Q3/5

True or false?

A vertical cut of a rectangular prism parallel to a side face gives a rectangle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S81-B1-Q4

Grade 7 • Geometry • 7.G.A.3 • Set 81 • BACKWARD 1 • Q4/5

True or false?

A cut of a cube parallel to a face can be a triangle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S81-B1-Q5

Grade 7 • Geometry • 7.G.A.3 • Set 81 • BACKWARD 1 • Q5/5

SET 81 • HSG-GMD.B.4

BACKWARD 2

Properties of polygons • Grade 5 • Geometry • 5.G.B.4

A quadrilateral with four right angles and unequal adjacent sides is a

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S81-B2-Q1

Grade 5 • Geometry • 5.G.B.4 • Set 81 • BACKWARD 2 • Q1/5

True or false?

Every square is a rectangle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S81-B2-Q2

Grade 5 • Geometry • 5.G.B.4 • Set 81 • BACKWARD 2 • Q2/5

A polygon with six sides is a

A. hexagon

B. pentagon

C. octagon

D. heptagon

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S81-B2-Q3

Grade 5 • Geometry • 5.G.B.4 • Set 81 • BACKWARD 2 • Q3/5

A triangle with two equal sides is

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S81-B2-Q4

Grade 5 • Geometry • 5.G.B.4 • Set 81 • BACKWARD 2 • Q4/5

True or false?

A trapezoid has no parallel sides.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S81-B2-Q5

Grade 5 • Geometry • 5.G.B.4 • Set 81 • BACKWARD 2 • Q5/5

SET 81 • HSG-GMD.B.4

BACKWARD 3

Parts of solids • Grade 6 • Geometry • 6.G.A.4

How many faces does a cube have?

Faces: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S81-B3-Q1

Grade 6 • Geometry • 6.G.A.4 • Set 81 • BACKWARD 3 • Q1/5

What is the shape of the base of a cone?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S81-B3-Q2

Grade 6 • Geometry • 6.G.A.4 • Set 81 • BACKWARD 3 • Q2/5

How many edges meet at each vertex of a cube?

A. 6

B. 2

C. 4

D. 3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S81-B3-Q3

Grade 6 • Geometry • 6.G.A.4 • Set 81 • BACKWARD 3 • Q3/5

True or false?

A cylinder has two congruent circular bases.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S81-B3-Q4

Grade 6 • Geometry • 6.G.A.4 • Set 81 • BACKWARD 3 • Q4/5

True or false?

A sphere has one edge.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S81-B3-Q5

Grade 6 • Geometry • 6.G.A.4 • Set 81 • BACKWARD 3 • Q5/5

SET 81 • HSG-GMD.B.4

FORWARD 1

Areas of cross-sections • High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4

A cube has edge 4. Find the area of the diagonal cross-section through two opposite edges.

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S81-F1-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 81 • FORWARD 1 • Q1/5

A cone has radius 6 and height 12. Find the area of the cross-section parallel to the base at height 4 (measured up from the base).

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S81-F1-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 81 • FORWARD 1 • Q2/5

True or false?

A sphere of radius 5 cut 3 units from its center gives a circle of radius 4.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S81-F1-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 81 • FORWARD 1 • Q3/5

A cube can have a cross-section that is a regular

A. octagon

B. pentagon

C. heptagon

D. hexagon

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S81-F1-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 81 • FORWARD 1 • Q4/5

A square pyramid has base side 10 and height 15. Find the area of the cross-section parallel to the base 5 units below the apex.

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S81-F1-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 81 • FORWARD 1 • Q5/5

SET 81 • HSG-GMD.B.4

FORWARD 2

Use cross-section areas to compare volumes (Cavalieri) • High School — Geometry, Advanced Extension (+) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.2

Two solids have height 10 and every horizontal cross-section of each has area 7. Compare their volumes.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S81-F2-Q1

High School — Geometry, Advanced Extension (+) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.2 • Set 81 • FORWARD 2 • Q1/5

A hemisphere has radius 5. Find the area of its cross-section 3 units above the base.

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S81-F2-Q2

High School — Geometry, Advanced Extension (+) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.2 • Set 81 • FORWARD 2 • Q2/5

True or false?

Cavalieri's principle can compare two solids of different heights.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S81-F2-Q3

High School — Geometry, Advanced Extension (+) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.2 • Set 81 • FORWARD 2 • Q3/5

At height y, a hemisphere of radius r has cross-section area

A. $\pi(r^2 - y^2)$

B. $\pi(r - y)^2$

C. πr^2

D. πy^2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S81-F2-Q4**

High School — Geometry, Advanced Extension (+) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.2 • Set 81 • FORWARD 2 • Q4/5

Explain why cross-section areas are useful for finding volumes.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S81-F2-Q5

High School — Geometry, Advanced Extension (+) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.2 • Set 81 • FORWARD 2 • Q5/5

SET 82 • HSG-GMD.B.4 • GEOMETRIC MEASUREMENT AND DIMENSION

Identify three-dimensional objects generated by rotations of two-dimensional objects

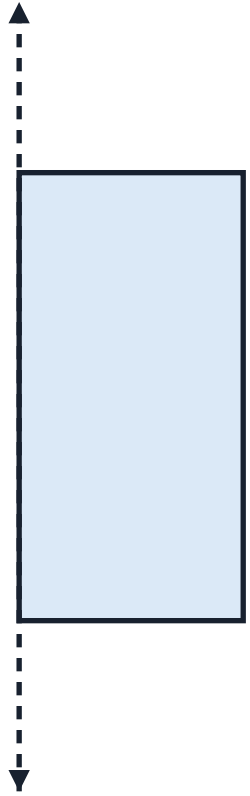
MAIN	Geometry main question	**HSG-GMD.B.4**
BACKWARD 1	Rotation of a point about a center	**HSG-CO.A.4**
BACKWARD 2	Cross-sections perpendicular to the axis	**HSG-GMD.B.4**
BACKWARD 3	Names of solids	**6.G.A.4**
FORWARD 1	Volumes of solids of rotation	**HSG-GMD.B.4**
FORWARD 2	Design an object as a solid of rotation	**HSG-MG.A.3**

SET 82 • HSG-GMD.B.4

MAIN

Geometry main question • High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4

The rectangle is rotated 360° about the dashed line. Name the solid formed.



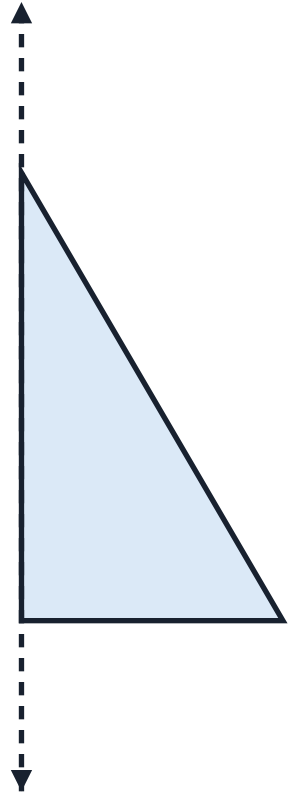
Solid: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S82-M-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 82 • MAIN • Q1/5

The right triangle is rotated 360° about the dashed line. Name the solid formed.



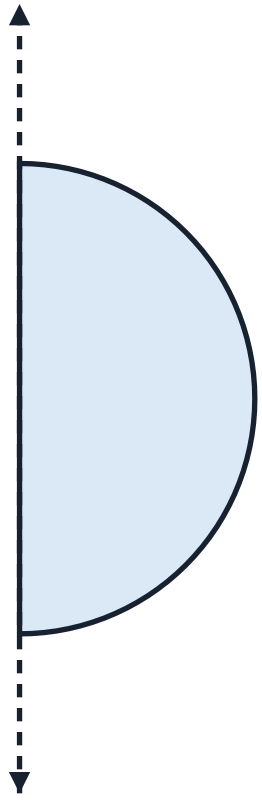
Solid: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S82-M-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 82 • MAIN • Q2/5

The semicircle is rotated 360° about the dashed line. Name the solid formed.



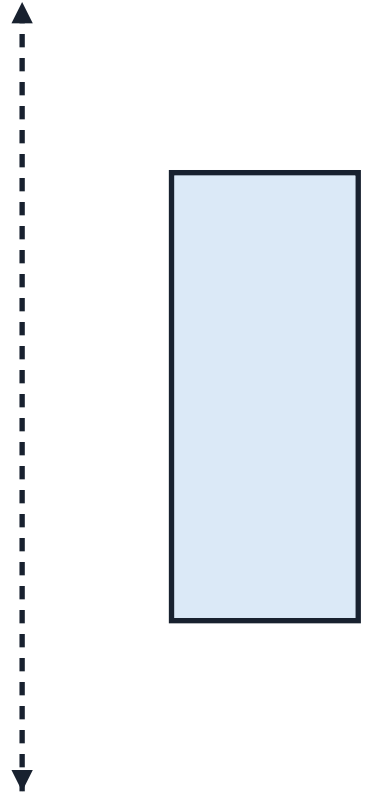
Solid: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S82-M-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 82 • MAIN • Q3/5

The rectangle is rotated 360° about the dashed line, which does not touch it. Describe the solid formed.



Solid: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S82-M-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 82 • MAIN • Q4/5

A right triangle is rotated about its hypotenuse. The solid formed is

A. a sphere

B. one cone

C. two cones joined at their bases

D. a cylinder

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S82-M-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 82 • MAIN • Q5/5

SET 82 • HSG-GMD.B.4

BACKWARD 1

Rotation of a point about a center • High School — Geometry • Geometry: Congruence • HSG-CO.A.4

True or false?

Rotating a point 360° about a center traces a circle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S82-B1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 82 • BACKWARD 1 • Q1/5

A point 3 cm from an axis is rotated about the axis. What does it trace?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S82-B1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 82 • BACKWARD 1 • Q2/5

A point ON the axis of rotation

A. traces a line

B. stays fixed

C. traces a circle

D. disappears

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S82-B1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 82 • BACKWARD 1 • Q3/5

A segment parallel to the axis, 2 units away, is rotated about it. What surface does it trace?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S82-B1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 82 • BACKWARD 1 • Q4/5

True or false?

Points farther from the axis trace smaller circles.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S82-B1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 82 • BACKWARD 1 • Q5/5

SET 82 • HSG-GMD.B.4

BACKWARD 2

Cross-sections perpendicular to the axis • High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4

True or false?

A solid of rotation cut perpendicular to its axis has a circular cross-section (or a ring).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S82-B2-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 82 • BACKWARD 2 • Q1/5

A cone is cut perpendicular to its axis. Shape?

Shape: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S82-B2-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 82 • BACKWARD 2 • Q2/5

A pipe cut perpendicular to its axis gives a

A. rectangle

B. ring (annulus)

C. circle

D. triangle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S82-B2-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 82 • BACKWARD 2 • Q3/5

A sphere is cut through its center. Shape?

Shape: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S82-B2-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 82 • BACKWARD 2 • Q4/5

True or false?

A cylinder cut perpendicular to its axis gives a rectangle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S82-B2-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 82 • BACKWARD 2 • Q5/5

SET 82 • HSG-GMD.B.4

BACKWARD 3

Names of solids • Grade 6 • Geometry • 6.G.A.4

Name the solid with two parallel circular bases and a curved side.

Solid: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S82-B3-Q1

Grade 6 • Geometry • 6.G.A.4 • Set 82 • BACKWARD 3 • Q1/5

Name the solid with one circular base and an apex.

Solid: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S82-B3-Q2

Grade 6 • Geometry • 6.G.A.4 • Set 82 • BACKWARD 3 • Q2/5

All points equidistant from a center in space form a

A. sphere

B. cone

C. circle

D. cylinder

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S82-B3-Q3

Grade 6 • Geometry • 6.G.A.4 • Set 82 • BACKWARD 3 • Q3/5

True or false?

A cone has one flat face.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S82-B3-Q4

Grade 6 • Geometry • 6.G.A.4 • Set 82 • BACKWARD 3 • Q4/5

True or false?

A sphere has a flat base.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S82-B3-Q5

Grade 6 • Geometry • 6.G.A.4 • Set 82 • BACKWARD 3 • Q5/5

SET 82 • HSG-GMD.B.4

FORWARD 1

Volumes of solids of rotation • High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4

A 3-by-5 rectangle is rotated about its 5-unit side. Find the exact volume of the solid.

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S82-F1-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 82 • FORWARD 1 • Q1/5

A 3-by-5 rectangle is rotated about its 3-unit side. Find the exact volume of the solid.

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S82-F1-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 82 • FORWARD 1 • Q2/5

True or false?

Rotating one rectangle about two different sides can give solids with different volumes.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S82-F1-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 82 • FORWARD 1 • Q3/5

A right triangle with legs 3 (on the axis) and 4 is rotated about the leg of length 3. Volume?

A. 48π

B. 9π

C. 12π

D. 16π

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S82-F1-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 82 • FORWARD 1 • Q4/5

A semicircle of radius 3 is rotated about its diameter. Find the exact volume.

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S82-F1-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 82 • FORWARD 1 • Q5/5

SET 82 • HSG-GMD.B.4

FORWARD 2

Design an object as a solid of rotation • High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3

A lathe turns a 2 cm by 10 cm rectangle about its 10 cm side to make a wooden dowel. Find the exact volume of wood.

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S82-F2-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 82 • FORWARD 2 • Q1/5

A designer wants a cone-shaped lampshade by rotating a right triangle about a 30 cm leg. What other leg length gives a base of diameter 40 cm?

Leg: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S82-F2-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 82 • FORWARD 2 • Q2/5

True or false?

A hemispherical bowl can be modeled by rotating a right triangle about one of its legs.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S82-F2-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 82 • FORWARD 2 • Q3/5

To make a ring-shaped washer by rotation, rotate a rectangle about a line that

A. is a diagonal

B. contains one side

C. does not touch the rectangle

D. passes through its center

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S82-F2-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 82 • FORWARD 2 • Q4/5

A vase is made by rotating a rectangle 6 cm wide and 20 cm tall about a line 2 cm from its long side (outside the rectangle). Find the exact volume of material.

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S82-F2-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 82 • FORWARD 2 • Q5/5

SET 83 • HSG-MG.A.1 • MODELING WITH GEOMETRY

Model objects with geometric shapes

MAIN	Geometry main question	**HSG-MG.A.1**
BACKWARD 1	Volume formulas	**HSG-GMD.A.3**
BACKWARD 2	Radius from a circumference	**7.G.B.4**
BACKWARD 3	Convert cubic units	**6.RP.A.3.d**
FORWARD 1	Judge a model against a measurement	**HSG-MG.A.1**
FORWARD 2	Use the model to estimate mass with density	**HSG-MG.A.2**

SET 83 • HSG-MG.A.1

MAIN

Geometry main question • High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1

A tree trunk is about 6 m tall and has circumference 2.2 m. Model it as a cylinder and estimate its volume to the nearest tenth of a cubic meter.

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S83-M-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 83 • MAIN • Q1/5

A pile of sand is shaped like a cone with base diameter 8 m and height 3 m. Estimate its volume to the nearest tenth of a cubic meter.

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S83-M-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 83 • MAIN • Q2/5

Which shape is the best model for a soup can?

A. Cone

B. Cylinder

C. Square pyramid

D. Sphere

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S83-M-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 83 • MAIN • Q3/5

A pool is modeled as a rectangular prism 25 m long, 10 m wide, and 1.8 m deep. How many liters of water does it hold? (1 m^3 = 1,000 L)

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S83-M-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 83 • MAIN • Q4/5

A pine tree is modeled as a cone. Name one measurement you would take and one way the model differs from the real tree.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S83-M-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 83 • MAIN • Q5/5

SET 83 • HSG-MG.A.1

BACKWARD 1

Volume formulas • High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3

Exact volume of a cylinder with r = 2 and h = 5?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S83-B1-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 83 • BACKWARD 1 • Q1/5

Exact volume of a cone with r = 3 and h = 4?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S83-B1-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 83 • BACKWARD 1 • Q2/5

Volume of a 2 × 3 × 4 prism?

A. 52

B. 9

C. 26

D. 24

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S83-B1-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 83 • BACKWARD 1 • Q3/5

True or false?

$V = \frac{1}{3}\pi r^2 h$ is the cone formula.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S83-B1-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 83 • BACKWARD 1 • Q4/5

True or false?

A sphere's volume is $\pi r^2 h$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S83-B1-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 83 • BACKWARD 1 • Q5/5

SET 83 • HSG-MG.A.1

BACKWARD 2

Radius from a circumference • Grade 7 • Geometry • 7.G.B.4

A circle has circumference 10π. Radius?

Radius: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S83-B2-Q1

Grade 7 • Geometry • 7.G.B.4 • Set 83 • BACKWARD 2 • Q1/5

A circle has circumference 2.2 m. Radius to the nearest hundredth?

Radius: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S83-B2-Q2

Grade 7 • Geometry • 7.G.B.4 • Set 83 • BACKWARD 2 • Q2/5

r =

A. C/(2π)

B. πC

C. 2πC

D. C/π

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S83-B2-Q3

Grade 7 • Geometry • 7.G.B.4 • Set 83 • BACKWARD 2 • Q3/5

True or false?

A tape measure around a trunk gives its circumference.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S83-B2-Q4

Grade 7 • Geometry • 7.G.B.4 • Set 83 • BACKWARD 2 • Q4/5

True or false?

The radius is C/2.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S83-B2-Q5

Grade 7 • Geometry • 7.G.B.4 • Set 83 • BACKWARD 2 • Q5/5

SET 83 • HSG-MG.A.1

BACKWARD 3

Convert cubic units • Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d

How many liters are in 2 m^3?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S83-B3-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d • Set 83 • BACKWARD 3 • Q1/5

How many cubic centimeters are in 1 L?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S83-B3-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d • Set 83 • BACKWARD 3 • Q2/5

450 m^3 = ___ L

A. 45,000

B. 0.45

C. 450,000

D. 4,500

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S83-B3-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d • Set 83 • BACKWARD 3 • Q3/5

True or false?

$1\ \text{mL} = 1\ \text{cm}^3$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S83-B3-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d • Set 83 • BACKWARD 3 • Q4/5

True or false?

$1\ m^3 = 100\ L$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S83-B3-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d • Set 83 • BACKWARD 3 • Q5/5

SET 83 • HSG-MG.A.1

FORWARD 1

Judge a model against a measurement • High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1

A water bottle is modeled as a cylinder with radius 3.8 cm and height 17 cm. The label says 750 mL. Find the model volume to the nearest mL and the percent error.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S83-F1-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 83 • FORWARD 1 • Q1/5

A silo is modeled as a cylinder (r = 3 m, h = 12 m) with a hemisphere on top. Find the exact volume.

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S83-F1-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 83 • FORWARD 1 • Q2/5

True or false?

A model that overestimates by 3% may still be useful for planning.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S83-F1-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 83 • FORWARD 1 • Q3/5

A football (pointed ends) is best modeled by

A. two cones joined at their bases

B. a cube

C. a sphere

D. a cylinder

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S83-F1-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 83 • FORWARD 1 • Q4/5

Why might a cylinder model overestimate the volume of a bottle?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S83-F1-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 83 • FORWARD 1 • Q5/5

SET 83 • HSG-MG.A.1

FORWARD 2

Use the model to estimate mass with density • High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2

A tree trunk modeled as a cylinder has volume about 2.3 m^3. The wood has density 600 kg/m^3. Estimate the trunk's mass to the nearest 10 kg.

Mass: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S83-F2-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 83 • FORWARD 2 • Q1/5

A sand pile modeled as a cone has volume 16π m³. The sand has density 1,600 kg/m³. Estimate its mass in metric tons (1 t = 1,000 kg), to the nearest ton.

Mass: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S83-F2-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 83 • FORWARD 2 • Q2/5

True or false?

A more accurate volume model gives a more accurate mass estimate.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S83-F2-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 83 • FORWARD 2 • Q3/5

A pool holds 450 m³ of water (1,000 kg/m³). Mass?

A. 450,000 kg

B. 450 kg

C. 45,000 kg

D. 4,500 kg

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S83-F2-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 83 • FORWARD 2 • Q4/5

Explain how a geometric model and a density work together to estimate the mass of a log.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S83-F2-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 83 • FORWARD 2 • Q5/5

SET 84 • HSG-MG.A.2 • MODELING WITH GEOMETRY

Apply density based on area

MAIN	Geometry main question	**HSG-MG.A.2**
BACKWARD 1	Unit rates	**6.RP.A.3.b**
BACKWARD 2	Areas of rectangles and circles	**7.G.B.6**
BACKWARD 3	Compare ratios	**7.RP.A.2**
FORWARD 1	Convert area units in density problems	**HSG-MG.A.2**
FORWARD 2	Plan capacity from an area density	**HSG-MG.A.3**

SET 84 • HSG-MG.A.2

MAIN

Geometry main question • High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2

A city has 840,000 people and an area of 120 square miles. Find its population density.

Density: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S84-M-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 84 • MAIN • Q1/5

A circular forest region has radius 3 km and 4,500 trees. Find the tree density to the nearest tree per square kilometer.

Density: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S84-M-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 84 • MAIN • Q2/5

Town A has 12,000 people on 30 km². Town B has 9,000 people on 20 km². Which town is more densely populated? Show the densities.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S84-M-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 84 • MAIN • Q3/5

A region has a population density of 250 people per km² and is a 4 km by 6 km rectangle. Estimate its population.

Population: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S84-M-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 84 • MAIN • Q4/5

Which is a unit of population density based on area?

A. people per mile

B. people per square mile

C. square miles per person per year

D. people

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S84-M-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 84 • MAIN • Q5/5

SET 84 • HSG-MG.A.2

BACKWARD 1

Unit rates • Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b

840 items in 12 boxes: items per box?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S84-B1-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 84 • BACKWARD 1 • Q1/5

$45 for 9 m² of tile: dollars per square meter?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S84-B1-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 84 • BACKWARD 1 • Q2/5

Unit rate for 300 people on 4 km²?

A. 0.013 km² per person

B. 1,200 people per km²

C. 296 people per km²

D. 75 people per km²

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S84-B1-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 84 • BACKWARD 1 • Q3/5

True or false?

A unit rate compares a quantity to one unit of another.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S84-B1-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 84 • BACKWARD 1 • Q4/5

True or false?

600 people on 3 km² is 1,800 people per km².

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S84-B1-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 84 • BACKWARD 1 • Q5/5

SET 84 • HSG-MG.A.2

BACKWARD 2

Areas of rectangles and circles • Grade 7 • Geometry • 7.G.B.6

Area of a 4 km by 6 km rectangle?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S84-B2-Q1

Grade 7 • Geometry • 7.G.B.6 • Set 84 • BACKWARD 2 • Q1/5

Exact area of a circle with radius 3 km?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S84-B2-Q2

Grade 7 • Geometry • 7.G.B.6 • Set 84 • BACKWARD 2 • Q2/5

A square region 5 mi on a side has area

A. 20 mi^2

B. 10 mi^2

C. 5 mi^2

D. 25 mi^2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S84-B2-Q3

Grade 7 • Geometry • 7.G.B.6 • Set 84 • BACKWARD 2 • Q3/5

True or false?

Area units are square units.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S84-B2-Q4

Grade 7 • Geometry • 7.G.B.6 • Set 84 • BACKWARD 2 • Q4/5

True or false?

A circle with radius 3 has area 6π.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S84-B2-Q5

Grade 7 • Geometry • 7.G.B.6 • Set 84 • BACKWARD 2 • Q5/5

SET 84 • HSG-MG.A.2

BACKWARD 3

Compare ratios • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2

Which is greater: 12,000/30 or 9,000/20?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S84-B3-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 84 • BACKWARD 3 • Q1/5

True or false?

To compare densities, compare the unit rates.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S84-B3-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 84 • BACKWARD 3 • Q2/5

Which rate is greatest?

A. 80 per 4 km²

B. 50 per 2 km²

C. 60 per 3 km²

D. 20 per 1 km²

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S84-B3-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 84 • BACKWARD 3 • Q3/5

A has 100 trees on 4 acres; B has 120 trees on 6 acres. Which is denser?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S84-B3-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 84 • BACKWARD 3 • Q4/5

True or false?

The town with more people is always denser.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S84-B3-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 84 • BACKWARD 3 • Q5/5

SET 84 • HSG-MG.A.2

FORWARD 1

Convert area units in density problems • High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2

1 square mile is about 2.59 km². Convert 7,000 people per square mile to people per km² (nearest whole number).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S84-F1-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 84 • FORWARD 1 • Q1/5

A lawn of 400 m² needs grass seed at 35 g per m². How many kilograms of seed are needed?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S84-F1-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 84 • FORWARD 1 • Q2/5

True or false?

A density in people per km² is a larger number than the same density in people per mi².

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S84-F1-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 84 • FORWARD 1 • Q3/5

A farm plants 2 trees per 10 m². How many trees on 1 hectare (10,000 m²)?

A. 2,000

B. 200

C. 5,000

D. 20,000

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S84-F1-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 84 • FORWARD 1 • Q4/5

A park of 3 hectares (1 ha = 10,000 m²) has 5,400 visitors. Visitors per 100 m²?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S84-F1-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 84 • FORWARD 1 • Q5/5

SET 84 • HSG-MG.A.2

FORWARD 2

Plan capacity from an area density • High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3

Safety rules allow 2 people per m^2 in a plaza. The plaza is a 30 m by 40 m rectangle. Maximum crowd?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S84-F2-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 84 • FORWARD 2 • Q1/5

A festival expects 5,000 people and wants at most 1.5 people per m². What is the smallest area needed (nearest m²)?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S84-F2-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 84 • FORWARD 2 • Q2/5

True or false?

When planning for a safety limit, round the capacity up.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S84-F2-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 84 • FORWARD 2 • Q3/5

A circular dance floor of radius 5 m allows 3 people per m². Capacity is about

A. 235 people

B. 471 people

C. 75 people

D. 47 people

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S84-F2-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 84 • FORWARD 2 • Q4/5

A circular dance floor of radius 5 m allows at most 3 people per m², and 3 · 25π ≈ 235.6. Explain why the capacity is 235 people, not 236.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S84-F2-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 84 • FORWARD 2 • Q5/5

SET 85 • HSG-MG.A.2 • MODELING WITH GEOMETRY

Apply density based on volume

MAIN	Geometry main question	**HSG-MG.A.2**
BACKWARD 1	Volume of prisms and cylinders	**HSG-GMD.A.3**
BACKWARD 2	Unit rates	**6.RP.A.3.b**
BACKWARD 3	Products of numbers in scientific notation	**8.EE.A.4**
FORWARD 1	Use density to identify materials and find volumes	**HSG-MG.A.2**
FORWARD 2	Design within a mass limit	**HSG-MG.A.3**

SET 85 • HSG-MG.A.2

MAIN

Geometry main question • High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2

An aluminum block measures 2 cm by 3 cm by 5 cm and has mass 81 g. Find its density.

Density: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S85-M-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 85 • MAIN • Q1/5

A cylindrical tank (radius 0.5 m, height 2 m) is full of water (1,000 kg/m³). Find the mass of the water to the nearest kilogram.

Mass: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S85-M-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 85 • MAIN • Q2/5

A heater must supply 20 BTU per cubic foot of room. A room is 12 ft by 15 ft with an 8 ft ceiling. How many BTU are needed?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S85-M-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 85 • MAIN • Q3/5

A sample has 3×10^6 bacteria per mL. How many bacteria are in 250 mL?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S85-M-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 85 • MAIN • Q4/5

Water has density 1 g/cm³. Which object floats in water?

A. A 1,000 g brick with volume 400 cm³

B. A 600 g block with volume 500 cm³

C. A 100 g ball with volume 50 cm³

D. A 400 g block with volume 500 cm³

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S85-M-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 85 • MAIN • Q5/5

SET 85 • HSG-MG.A.2

BACKWARD 1

Volume of prisms and cylinders • High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3

Volume of a 2 × 3 × 5 cm block?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S85-B1-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 85 • BACKWARD 1 • Q1/5

Exact volume of a cylinder with r = 0.5 m, h = 2 m?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S85-B1-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 85 • BACKWARD 1 • Q2/5

Volume of a 12 × 15 × 8 ft room?

A. 1,440 ft^3

B. 180 ft^3

C. 2,880 ft^3

D. 35 ft^3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S85-B1-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 85 • BACKWARD 1 • Q3/5

True or false?

Volume is needed before a volume density can be used.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S85-B1-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 85 • BACKWARD 1 • Q4/5

True or false?

A 1 m cube has volume 3 m³.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S85-B1-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 85 • BACKWARD 1 • Q5/5

SET 85 • HSG-MG.A.2

BACKWARD 2

Unit rates • Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b

81 g in 30 cm³: grams per cm³?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S85-B2-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 85 • BACKWARD 2 • Q1/5

20 BTU per ft³ for 100 ft³: total?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S85-B2-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 85 • BACKWARD 2 • Q2/5

500 g in 250 cm³ is

A. 250 g/cm³

B. 2 g/cm³

C. 0.5 g/cm³

D. 750 g/cm³

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S85-B2-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 85 • BACKWARD 2 • Q3/5

True or false?

Amount = rate × quantity.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S85-B2-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 85 • BACKWARD 2 • Q4/5

True or false?

Density = volume ÷ mass.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S85-B2-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 85 • BACKWARD 2 • Q5/5

SET 85 • HSG-MG.A.2

BACKWARD 3

Products of numbers in scientific notation • Grade 8 • Expressions & Equations • 8.EE.A.4

$(3 \times 10^6) \times 250$ in scientific notation?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S85-B3-Q1

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 85 • BACKWARD 3 • Q1/5

Write 2,400,000 in scientific notation.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S85-B3-Q2

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 85 • BACKWARD 3 • Q2/5

$(2 \times 10^3)(4 \times 10^2) =$

A. 6×10^5

B. 8×10^5

C. 8×10^6

D. 8×10^1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S85-B3-Q3

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 85 • BACKWARD 3 • Q3/5

True or false?

$10^3 \times 10^2 = 10^5$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S85-B3-Q4

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 85 • BACKWARD 3 • Q4/5

True or false?

$5 \times 10^2 \times 3 = 15 \times 10^6$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S85-B3-Q5

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 85 • BACKWARD 3 • Q5/5

SET 85 • HSG-MG.A.2

FORWARD 1

Use density to identify materials and find volumes • High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2

A ring has mass 9.65 g and volume 0.5 cm³. Gold has density 19.3 g/cm³. Could the ring be pure gold?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S85-F1-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 85 • FORWARD 1 • Q1/5

A 1,350 g piece of aluminum (2.7 g/cm³) is melted into a cube. Find the edge of the cube, to the nearest tenth of a centimeter.

Edge: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S85-F1-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 85 • FORWARD 1 • Q2/5

True or false?

Two objects with equal mass but different volumes have different densities.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S85-F1-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 85 • FORWARD 1 • Q3/5

Oil (0.9 g/cm³) in a 2,000 cm³ jug has mass

A. 1,800 g

B. 2,000 g

C. 2,222 g

D. 1,100 g

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S85-F1-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 85 • FORWARD 1 • Q4/5

A balloon holds 0.02 m³ of helium (0.18 kg/m³). Mass of the helium in grams?

Mass: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S85-F1-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 85 • FORWARD 1 • Q5/5

SET 85 • HSG-MG.A.2

FORWARD 2

Design within a mass limit • High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3

A truck can carry 12,000 kg. Gravel has density 1,500 kg/m³. What is the largest volume of gravel it can carry?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S85-F2-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 85 • FORWARD 2 • Q1/5

A shelf holds at most 40 kg. Books have density 0.8 g/cm³. How many cubic centimeters of books can it hold?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S85-F2-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 85 • FORWARD 2 • Q2/5

True or false?

Mass limits become volume limits by multiplying by the density.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S85-F2-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 85 • FORWARD 2 • Q3/5

A cylindrical water tank must weigh at most 3,140 kg when full (water 1,000 kg/m³, ignore the tank). With radius 1 m, its height can be at most about

A. 10 m

B. 3.14 m

C. 1 m

D. 0.32 m

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S85-F2-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 85 • FORWARD 2 • Q4/5

A concrete block must be under 25 kg. Concrete is 2,400 kg/m³. Will a 20 cm × 20 cm × 30 cm block work?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S85-F2-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 85 • FORWARD 2 • Q5/5

SET 86 • HSG-MG.A.3 • MODELING WITH GEOMETRY

Design to satisfy physical constraints

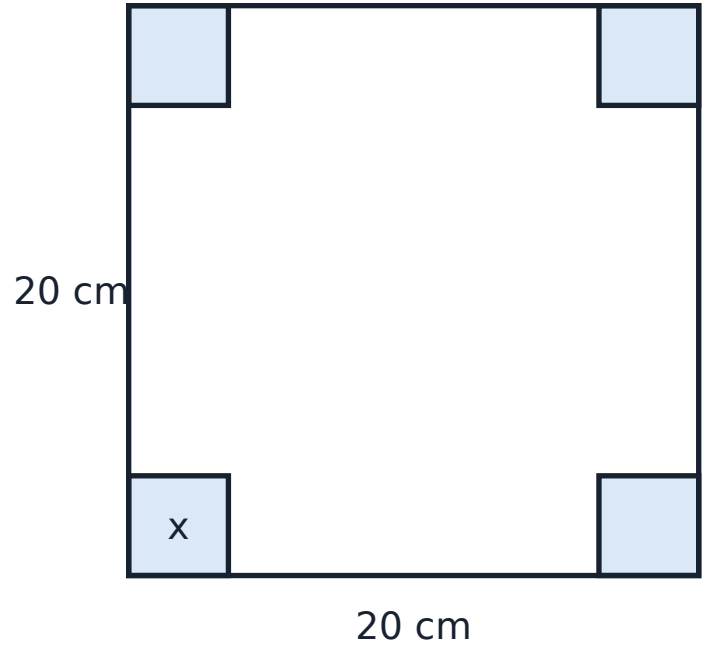
MAIN	Geometry main question	**HSG-MG.A.3**
BACKWARD 1	Volume of a rectangular prism	**7.G.B.6**
BACKWARD 2	Use the Pythagorean Theorem	**8.G.B.7**
BACKWARD 3	Ratios and rates	**7.RP.A.2**
BACKWARD 4	Inequalities as limits	**7.EE.B.4**
FORWARD 1	Satisfy several constraints at once	**HSG-MG.A.3**
FORWARD 2	Find the best design with a function graph	**HSF-IF.C.7.c**

SET 86 • HSG-MG.A.3

MAIN

Geometry main question • High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3

An open box is made from a 20 cm by 20 cm sheet by cutting an x-cm square from each corner and folding up the sides. Compare the volumes for x = 2, 3, and 4. Which gives the largest box?



Not drawn to scale.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S86-M-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 86 • MAIN • Q1/5

Accessibility rules allow a ramp slope of at most 1 : 12 (1 unit of rise per 12 units of run). A ramp must rise 30 in. What is the shortest horizontal run allowed, in feet?

Run: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S86-M-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 86 • MAIN • Q2/5

A rectangular pen will use 40 m of fencing. Compare a 10 m by 10 m pen with a 5 m by 15 m pen. Which encloses more area?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S86-M-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 86 • MAIN • Q3/5

A cylindrical tank must fit in a closet 2 m wide, 2 m deep, and 3 m tall. What is the largest possible volume, to the nearest tenth of a cubic meter?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S86-M-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 86 • MAIN • Q4/5

A 13 ft ladder leans on a wall. For safety its base must be at least 5 ft from the wall. The highest point it can reach is

A. 18 ft

B. 12 ft

C. 13 ft

D. 8 ft

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S86-M-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 86 • MAIN • Q5/5

SET 86 • HSG-MG.A.3

BACKWARD 1

Volume of a rectangular prism • Grade 7 • Geometry • 7.G.B.6

Volume of a box 14 cm by 14 cm by 3 cm?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S86-B1-Q1

Grade 7 • Geometry • 7.G.B.6 • Set 86 • BACKWARD 1 • Q1/5

A 20 cm side with 3 cm cut from each end leaves

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S86-B1-Q2

Grade 7 • Geometry • 7.G.B.6 • Set 86 • BACKWARD 1 • Q2/5

Volume of a box 12 by 12 by 4?

A. 28

B. 576

C. 48

D. 144

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S86-B1-Q3

Grade 7 • Geometry • 7.G.B.6 • Set 86 • BACKWARD 1 • Q3/5

True or false?

Cutting larger corners makes the box taller but its base smaller.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S86-B1-Q4

Grade 7 • Geometry • 7.G.B.6 • Set 86 • BACKWARD 1 • Q4/5

True or false?

An open box is made from a 20 cm by 20 cm sheet by cutting an x-cm square from each corner. The box's base side is $20 - x$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S86-B1-Q5

Grade 7 • Geometry • 7.G.B.6 • Set 86 • BACKWARD 1 • Q5/5

SET 86 • HSG-MG.A.3

BACKWARD 2

Use the Pythagorean Theorem • Grade 8 • Geometry • 8.G.B.7

Hypotenuse 13, leg 5: other leg?

Leg: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S86-B2-Q1

Grade 8 • Geometry • 8.G.B.7 • Set 86 • BACKWARD 2 • Q1/5

Legs 30 in. and 360 in.: hypotenuse to the nearest inch?

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S86-B2-Q2

Grade 8 • Geometry • 8.G.B.7 • Set 86 • BACKWARD 2 • Q2/5

A ladder, wall, and ground form a

A. right triangle

B. circle

C. square

D. equilateral triangle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S86-B2-Q3

Grade 8 • Geometry • 8.G.B.7 • Set 86 • BACKWARD 2 • Q3/5

True or false?

A ladder leans against a vertical wall on level ground. The ladder is the hypotenuse of the right triangle formed.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S86-B2-Q4

Grade 8 • Geometry • 8.G.B.7 • Set 86 • BACKWARD 2 • Q4/5

True or false?

The ladder's height on the wall is 13 + 5.

A. True

B. False

SET 86 • HSG-MG.A.3

BACKWARD 3

Ratios and rates • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2

A ratio of 1 : 12 for a rise of 2 in. gives what run?

Run: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S86-B3-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 86 • BACKWARD 3 • Q1/5

Is a slope of 3 : 30 steeper than 1 : 12?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S86-B3-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 86 • BACKWARD 3 • Q2/5

360 in. =

A. 30 ft

B. 36 ft

C. 12 ft

D. 3.6 ft

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S86-B3-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 86 • BACKWARD 3 • Q3/5

True or false?

A ratio of 1 : 12 means 1 inch up for every 12 inches across.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S86-B3-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 86 • BACKWARD 3 • Q4/5

True or false?

A 1 : 8 ramp is less steep than a 1 : 12 ramp.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S86-B3-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 86 • BACKWARD 3 • Q5/5

SET 86 • HSG-MG.A.3

BACKWARD 4

Inequalities as limits • Grade 7 • Expressions & Equations • 7.EE.B.4

The run r must satisfy r ≥ 360. Is r = 300 allowed?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S86-B4-Q1

Grade 7 • Expressions & Equations • 7.EE.B.4 • Set 86 • BACKWARD 4 • Q1/5

A diameter d must satisfy d ≤ 2. Largest radius?

Radius: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S86-B4-Q2

Grade 7 • Expressions & Equations • 7.EE.B.4 • Set 86 • BACKWARD 4 • Q2/5

"At most 3 m" means

A. ≥ 3

B. < 3 only

C. ≤ 3

D. = 3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S86-B4-Q3

Grade 7 • Expressions & Equations • 7.EE.B.4 • Set 86 • BACKWARD 4 • Q3/5

True or false?

"At least 5 ft" allows exactly 5 ft.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S86-B4-Q4

Grade 7 • Expressions & Equations • 7.EE.B.4 • Set 86 • BACKWARD 4 • Q4/5